

MAH-CET 2025

Slot - 4

with Solution

Section 1 Logic

Q1

Look carefully for the pattern, and then choose which pair of numbers comes next.

28 25 5 21 18 5 14

A-11,5

B-10,7

C-11,8

D-5,10

E-10,5

Q2

Look carefully for the pattern, and then choose which pair of numbers comes next.

13 29 15 26 17 23 19

A-21,23

B-20,21

C-20,17

D-25,27

E-22,20

Q3 - 07

Based on the criteria and the information provided below, assess the given candidate's eligibility or potential and thereby decide the action to be taken from among the given alternatives.

A public charitable trust desires to select Medical Officers for its rural hospital based on the following criteria:

The applicant must -

(i) be holding MBBS degree with a minimum of 50% marks

(ii) have a minimum of 4 years of experience of full-time practice in rural areas

(iii) be ready to execute a bond of 3 years of service

(iv) have good knowledge of the local language

In the case of the applicant who satisfies all other criteria except -

(A) at (ii) above, but has 4 years of full-time experience of practice either urban or semi-urban areas and spent at least 5 years in rural areas any time during his life, be referred to Secretary of the Trust

(B) at (iv) above, but has working knowledge of Hindi be referred to Assistant Secretary of the Trust.

(C) at (i) above, but has a minimum of 45% marks at MBBS and has done M.S. or M.D., with a minimum of 50%, be referred to the Chief Medical Officer (C.M.O.)

(D) at (iii) above, but is ready to give Rs. 25,000 as security money, be referred to the President of the Trust.

Q3

Case: Gauri did her M.D. after doing her MBBS. She is ready to execute three years bond of service. She has good command over local language as well as Hindi. She has practised for 5½ years in remote village

out of her love for social service. She has obtained 77%, 88%, 47% and 56% at SS MBBS and M.D. respectively.

A-To be selected

B-Not to be selected

C-Refer to Assistant Secretary

D-Data inadequate

E-Refer to the CMO

Q4

Case: Payal, after obtaining her MBBS and M.S., decided to practice in her native village for five years. She knows the local language very well. Her dispensary and small hospital were very popular in the nearby villages. She plans to go to U.S.A. and U.K. after spending 4 more years in India. She has secured more than 60% marks in all the examinations right from SSC to M.S. She is ready to execute a bond of 3 years of service.

A-To be selected

B-Not to be selected

C-Refer to Assistant Secretary

D-Data inadequate

E-Refer to the CMO

Q5

Case: Raghav studied in rural areas while doing his schooling. His father is a farmer. He completed his MBBS in Mumbai and had six years of practice in a big city. He has good knowledge of the local language and working knowledge of Hindi. He is ready to execute a 3-year bond of service. He has done M.S. with 53% marks.

A-To be selected

B-Not to be selected

C-Refer to Assistant Secretary

D-Data inadequate

E-Refer to the CMO

Q6

In the question given below, two statements are given, followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusion and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

Statements: No women teacher can play. Some women teachers are athletes.

Conclusions:

1. Male athletes can play.
2. Some athletes can play.

- A-Only 1 follows
- B-Only 2 follows
- C-Either 1 or 2 follows
- D-Neither 1 nor 2 follows
- E-Both 1 and 2 follow

Q7

In the question given below, two statements are given, followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusion and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

Statements: All mangoes are golden in colour. No golden-coloured things are cheap.

Conclusions:

1. All mangoes are cheap.
2. Golden-coloured mangoes are not cheap.

- A-Only 1 follows
- B-Only 2 follows
- C-Either 1 or 2 follows

- D-Neither 1 nor 2 follows
E-Both 1 and 2 follow



Q8

In the question given below, two statements are given, followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusion and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

Statements: All flowers are trees. No fruit is tree.

Conclusions:

1. No fruit is flower.
2. Some trees are flowers.

- A-Only 1 follows
B-Only 2 follows
C-Either 1 or 2 follows
D-Neither 1 nor 2 follows
E-Both 1 and 2 follow

Q9

In the question given below, two statements are given, followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusion and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

Statements: Some swords are sharp. All swords are rusty.

Conclusions:

1. Some rusty things are sharp.
2. Some rusty things are not sharp.

- A-Only 1 follows

- B-Only 2 follows
C-Either 1 or 2 follows
D-Neither 1 nor 2 follows
E-Both 1 and 2 follow



Q10

Look at this series: 8, 43, 11, 41, __, 39, 17, ... What number should fill in the blank?

- A-13
B-15
C-14
D-16
E-40

Q11

Given below is a statement and two assumptions. These assumptions may or may not be implicit in the statement.

Statement: The best evidence of India's glorious past is the growing popularity of ayurvedic medicines in the world.

Assumption A: Ayurvedic medicines are popular in India.

Assumption B: Ayurvedic medicines are not popular in India.

Determine whether one or both assumptions are implicit or not in the given statement and choose the correct option from the options given below:

- A-Only Assumption A is implicit
B-Only Assumption B is implicit
C-Neither Assumption A nor Assumption B is implicit
D-Both Assumption A and Assumption B are implicit
E-None of these

Q12



Given below is a statement and two assumptions. These assumptions may or may not be implicit in the statement.

Statement: "In order to bring punctuality in our office, we must provide conveyance allowance to our employees." - In charge of a company tells Personnel Manager.

Assumption A: Conveyance allowance will not help in bringing punctuality.

Assumption B: Discipline and reward should always go hand in hand.

Determine whether one or both assumptions are implicit or not in the given statement and choose the correct option from the options given below:

- A-Only Assumption A is implicit
- B-Only Assumption B is implicit
- C-Neither Assumption A nor Assumption B is implicit
- D-Both Assumption A and Assumption B are implicit
- E-None of these

Q13

Given below is a statement and two assumptions. These assumptions may or may not be implicit in the statement.

Statement: The State government has decided to appoint four thousand primary school teachers during the next financial year.

Assumption A: There are enough schools in the state to accommodate four thousand additional primary school teachers.

Assumption B: The eligible candidates may not be interested to apply as the government may not finally appoint such a large number of primary school teachers.

Determine whether one or both assumptions are implicit or not in the given statement and choose the correct option from the options given below:

- A-Only Assumption A is implicit
- B-Only Assumption B is implicit
- C-Neither Assumption A nor Assumption B is implicit

D-Both Assumption A and Assumption B are implicit
E-None of these



Q14

In a row at a bus stop, 'A' is 7th from the left and 'B' is 9th from the right.
They both interchange their positions. Now A becomes 11th from the left.
How many people are there in the row?

- A-15
- B-16
- C-17
- D-19
- E-21

Q15

Complete the series by filling the blank with suitable alternative.

CMM, EOO, GQQ, _____, KUU

- A-GRR
- B-GSS
- C-ISS
- D-ITT
- E-HSS

Q16

Find the next term in the series

4, 6, 12, 14, 28, 30, _____

- A-44
- B-60
- C-52
- D-32
- E-40

Q17

Complete the series

10000, 11000, 9900, 10890, 9801, ?

- A-10781
- B-10241
- C-10423
- D-10929

E-10900



Q18

Choose the number pair/group which is different from others.

- A- 70-80
- B- 54-62
- C- 28-32
- D- 21-24
- E- 14-16

Q19

Choose the number group from the given alternatives which is different from others.

- A- 71,7,3,17
- B- 67,71,3,5
- C- 41,5,3,47
- D- 37,14,19,7
- E- 11,3,5,17

Q20

If ROAST is coded as PQYUR in a certain language, then how shall SLOPPY be coded in that Language?

- A-MRNAQN
- B-NRMNQA
- C-QNMRNA
- D-RANNMQ
- E-RANMNQ

Q21

In a certain language if "health" is written as "GSKZDG", then "north" is written as ?

- A-GSQNM
- B-GSQMN
- C-FSQNM
- D-FSQMN
- E-GSRNM

Q22

In a certain code language, BEAT is written as YVZG, then what will be the code of MILD?

- A-ONRW

B-OWNR
C-ONWR
D-NROW
E-NORW



Q23

Directions: Study the following information carefully and answer the question that follows:

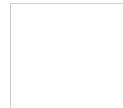
(i) A, B, C, D, E, F and G are seven persons wearing shirts of different colors-white, red, black, green, yellow, blue and violet-and trousers of different colors-blue, red, white, black, cream, yellow and indigo. The persons, color of the shirt and color of the trousers above are not necessarily in the same order. No person is wearing shirt and trousers of same color.

(ii) B is wearing red shirt and is not wearing cream or yellow trousers. D is wearing green shirt and indigo trousers. The color of A's shirt and F's trousers is same. The color of E's shirt and C's trousers is same. G is wearing blue shirt and E is wearing blue trousers. F is not wearing any yellow dress. A is not wearing a white shirt. Red and blue is not the combination of shirt and trousers of any of the persons.

What is the colour of G's trousers?

A-White
B-Cream
C-Green
D-Indigo
E-Red

Q24



Directions: Study the following information carefully and answer the question that follows:

(i) A, B, C, D, E, F and G are seven persons wearing shirts of different colors-white, red, black, green, yellow, blue and violet-and trousers of different colors-blue, red, white, black, cream, yellow and indigo. The persons, color of the shirt and color of the trousers above are not necessarily in the same order. No person is wearing shirt and trousers of same color.

(ii) B is wearing red shirt and is not wearing cream or yellow trousers. D is wearing green shirt and indigo trousers. The color of A's shirt and F's trousers is same. The color of E's shirt and C's trousers is same. G is wearing blue shirt and E is wearing blue trousers. F is not wearing any yellow dress. A is not wearing a white shirt. Red and blue is not the combination of shirt and trousers of any of the persons.

Who wears a violet colored shirt?

- A-C
- B-F
- C-C or F
- D-G
- E-C or G

Q25



Directions: Study the following information carefully and answer the question that follows:

(i) A, B, C, D, E, F and G are seven persons wearing shirts of different colors-white, red, black, green, yellow, blue and violet-and trousers of different colors-blue, red, white, black, cream, yellow and indigo. The persons, color of the shirt and color of the trousers above are not necessarily in the same order. No person is wearing shirt and trousers of same color.

(ii) B is wearing red shirt and is not wearing cream or yellow trousers. D is wearing green shirt and indigo trousers. The color of A's shirt and F's trousers is same. The color of E's shirt and C's trousers is same. G is wearing blue shirt and E is wearing blue trousers. F is not wearing any yellow dress. A is not wearing a white shirt. Red and blue is not the combination of shirt and trousers of any of the persons.

What is the color of B's trouser?

- A-Black
- B-White
- C-Indigo
- D-Blue
- E-Yellow

Q26

Here are some words translated from an artificial language.

agnoscrenia means poisonous spider

delanocrenia means poisonous snake

agnosdeery means brown spider

Which word could mean "black widow spider"?

- A-deeryclostagnos
- B-agnosdelano
- C-agnosvitribulunin
- D-trymuttiagnos
- E-deerydelano

Q27



Read the following information carefully and answer the question that follows:

Sumeet, Philips, Wasim, Bishan and Chetan are five players of the College Cricket Team and their hometowns are Surat, Pune, Warangal, Bangalore and Chandigarh but not in that order. The five specialist slots of spinner, pace bowler, wicket keeper, batsman and captain are held by them, again not in the order of their names stated above.

- (i) Their names, hometowns and specialities do not start with the same letter.
- (ii) Neither Philips nor Wasim is the captain and they do not belong to either Surat or Bangalore.
- (iii) Sumeet is neither a wicket keeper nor a batsman.
- (iv) Pune is not Bishan's hometown.
- (v) The player who hails from Bangalore is a wicket keeper.
- (vi) The captain's hometown is Pune while the batsman does not hail from Warangal.

Who is the spinner ?

- A-Philips
- B-Chetan
- C-Bishan
- D-Wasim
- E-Sumeet

Q28

Read the following information carefully and answer the question that follows:

Sumeet, Philips, Wasim, Bishan and Chetan are five players of the College Cricket Team and their hometowns are Surat, Pune, Warangal, Bangalore and Chandigarh but not in that order. The five specialist slots of spinner, pace bowler, wicket keeper, batsman and captain are held by them, again not in the order of their names stated above.

- (i) Their names, hometowns and specialities do not start with the same letter.
- (ii) Neither Philips nor Wasim is the captain and they do not belong to either Surat or Bangalore.
- (iii) Sumeet is neither a wicket keeper nor a batsman.
- (iv) Pune is not Bishan's hometown.
- (v) The player who hails from Bangalore is a wicket keeper.
- (vi) The captain's hometown is Pune while the batsman does not hail from Warangal.

The Spinner's hometown is?

- A-Chandigarh
- B-Pune
- C-Warangal
- D-Surat
- E-Bangalore

Read the following information carefully and answer the question that follows:

Sumeet, Philips, Wasim, Bishan and Chetan are five players of the College Cricket Team and their hometowns are Surat, Pune, Warangal, Bangalore and Chandigarh but not in that order. The five specialist slots of spinner, pace bowler, wicket keeper, batsman and captain are held by them, again not in the order of their names stated above.

- (i) Their names, hometowns and specialities do not start with the same letter.
- (ii) Neither Philips nor Wasim is the captain and they do not belong to either Surat or Bangalore.
- (iii) Sumeet is neither a wicket keeper nor a batsman.
- (iv) Pune is not Bishan's hometown.
- (v) The player who hails from Bangalore is a wicket keeper.
- (vi) The captain's hometown is Pune while the batsman does not hail from Warangal.

Chandigarh is the hometown of?

- A-Sumeet
- B-Wasim
- C-Philips
- D-Bishan
- E-Chetan

Q30

Read the following information carefully and answer the question that follows:

Sumeet, Philips, Wasim, Bishan and Chetan are five players of the College Cricket Team and their hometowns are Surat, Pune, Warangal, Bangalore and Chandigarh but not in that order. The five specialist slots of spinner, pace bowler, wicket keeper, batsman and captain are held by them, again not in the order of their names stated above.

- (i) Their names, hometowns and specialities do not start with the same letter.
- (ii) Neither Philips nor Wasim is the captain and they do not belong to either Surat or Bangalore.
- (iii) Sumeet is neither a wicket keeper nor a batsman.
- (iv) Pune is not Bishan's hometown.
- (v) The player who hails from Bangalore is a wicket keeper.
- (vi) The captain's hometown is Pune while the batsman does not hail from Warangal.

Who is the pace-bowler?

- A-Chetan
- B-Wasim
- C-Sumeet
- D-Bishan
- E-Philips

Q31

Here are some words translated from an artificial language.

migenlasan means cupboard

lasanpoen means boardwalk

cuopdansa means pullman

Which word could mean "walkway"?

- A-poenmigen
- B-cuopeisel
- C-lasandansa
- D-poenforc
- E-lasanforc

Q32

Study the following information and answer the question that follows:

A blacksmith has five iron articles A, B, C, D, and E, each having a different weight.

- 1) A weighs twice as much as B.
- 2) B weighs four-and-a-half times as much as C.
- 3) C weighs half as much as D.
- 4) D weighs half as much as E.
- 5) E weighs less than A but more than C.

Which of the following is the lightest in weight?

- A-A
- B-B
- C-C
- D-D
- E-E

Q33

Study the following information and answer the question that follows:

A blacksmith has five iron articles A, B, C, D, and E, each having a different weight.

- 1) A weighs twice as much as B.
- 2) B weighs four-and-a-half times as much as C.
- 3) C weighs half as much as D.
- 4) D weighs half as much as E.
- 5) E weighs less than A but more than C.

E is lighter than which of the following two articles?

- A- A,B
- B- B,C
- C- A,D
- D- A,C
- E- B,D

Q34

Study the following information and answer the question that follows:

A blacksmith has five iron articles A, B, C, D, and E, each having a different weight.

- 1) A weighs twice as much as B.
- 2) B weighs four-and-a-half times as much as C.
- 3) C weighs half as much as D.
- 4) D weighs half as much as E.
- 5) E weighs less than A but more than C.

E is heavier than which of the following two articles?

- A- A,C
- B- D,C
- C- B,C
- D- A,B
- E- A,C

Q35

If it is possible to make a meaningful word from the 1st, 3rd, 6th, and 8th letters of the word "NOMINATION", then which will be the 3rd letter of the word? Mark X if no such word can be formed, Y if more than one such word can be formed.

- A-N
- B-I
- C-M
- D-X
- E-Y

Q36

Study the following information carefully and answer the question that follows:

A, B, C, E, F, G and H are seven employees in an organization working in the departments of Administration, Accounts and Operations. There are at least two employees in each department. There are three females, one in each department. Each of the seven employees earns a different amount. F works in Administration and his only other colleague, G earns the maximum. C, the least earner, works in Accounts. B and E are brothers and do not work in the same department. A, husband of H, works in Accounts and earns more than each of F, B and E. The wife in the couple earns more than the husband.

In which department do three people work?

- A-Operations
- B-Accounts
- C-Administration
- D-Accounts and Administration
- E-Operation or Accounts

Q37

Study the following information carefully and answer the question that follows:

A, B, C, E, F, G and H are seven employees in an organization working in the departments of Administration, Accounts and Operations. There are at least two employees in each department. There are three females, one in each department. Each of the seven employees earns a different amount. F works in Administration and his only other colleague, G earns the maximum. C, the least earner, works in Accounts. B and E are brothers and do not work in the same department. A, husband of H, works in Accounts and earns more than each of F, B and E. The wife in the couple earns more than the husband.

Which of the following is a group of females?

A-G,E,H
B-G,C,H
C-G,C,E
D-G,H,B
E-G,H,F

Q38

Study the following information carefully and answer the question that follows:

A, B, C, E, F, G and H are seven employees in an organization working in the departments of Administration, Accounts and Operations. There are at least two employees in each department. There are three females, one in each department. Each of the seven employees earns a different amount. F works in Administration and his only other colleague, G earns the maximum. C, the least earner, works in Accounts. B and E are brothers and do not work in the same department. A, husband of H, works in Accounts and earns more than each of F, B and E. The wife in the couple earns more than the husband.

What will be the position of A from the top when they are arranged in descending order of income?

- A- First
- B- Second
- C- Third
- D- Fourth
- E- Fifth

Q39

Find the missing number.

72	24	6
96	16	12
108	?	18

- A-24
- B-12
- C-18
- D-30

E-6



Q40

In a club of 30 people, all of them belong to at least one group - Chess, Drama and Art. 6 people belong only to the Art group. 5 people belong to all three groups. 2 people have joined the Chess and the Art group but not the Drama group. 15 people belong to the Art group. 2 people have joined only the Chess group. 3 people have joined only the Drama group.

How many people belong to exactly one group?

A-11

B-9

C-15

D-14

E-13

Q41

Study the following information and answer the question that follows:

Seven specialist doctors, P, Q, R, S, T, U and V, visit a polyclinic on any one of the four days in a week, i.e. on Tuesday, Wednesday, Friday or Saturday.

At least one doctor but not more than two doctors visit the polyclinic on each of these days. Each of them is a specialist in different fields - ENT, Orthopaedics, Paediatrics, Neurology, ophthalmology, Radiology and Oncology.

S visits on Friday with Radiologist, Paediatrician does not visit on Saturday nor with T and V.

Oncologist U visits alone on Tuesday.

Q visits on Wednesday and he is not a Paediatrician.

R visits on Wednesday.

V is not a Radiologist.

Paediatrician visits with ENT specialist.

Neurologist visits on Friday.

P is neither Orthopaedics nor Radiologist.

On which of the following days do the specialists in Orthopaedic and Ophthalmology visit?

- A-Tuesday
- B-Wednesday
- C-Friday
- D-Saturday
- E-Tuesday and Saturday

Q42

Study the following information and answer the question that follows:

Seven specialist doctors, P, Q, R, S, T, U and V, visit a polyclinic on any one of the four days in a week, i.e. on Tuesday, Wednesday, Friday or Saturday.

At least one doctor but not more than two doctors visit the polyclinic on each of these days. Each of them is a specialist in different fields - ENT, Orthopaedics, Paediatrics, Neurology, ophthalmology, Radiology and Oncology.

S visits on Friday with Radiologist, Paediatrician does not visit on Saturday nor with T and V.

Oncologist U visits alone on Tuesday.

Q visits on Wednesday and he is not a Paediatrician.

R visits on Wednesday.

V is not a Radiologist.



Paediatrician visits with ENT specialist.

Neurologist visits on Friday.

P is neither Orthopaedics nor Radiologist.

What is the profession of S?

- A-Oncologist
- B-Neurologist
- C-Radiologist
- D-Ophthalmologist
- E-ENT

Q43

Study the following information and answer the question that follows:

Seven specialist doctors, P, Q, R, S, T, U and V, visit a polyclinic on any one of the four days in a week, i.e. on Tuesday, Wednesday, Friday or Saturday.

At least one doctor but not more than two doctors visit the polyclinic on each of these days. Each of them is a specialist in different fields - ENT, Orthopaedics, Paediatrics, Neurology, ophthalmology, Radiology and Oncology.

S visits on Friday with Radiologist, Paediatrician does not visit on Saturday nor with T and V.

Oncologist U visits alone on Tuesday.

Q visits on Wednesday and he is not a Paediatrician.

R visits on Wednesday.

V is not a Radiologist.

Paediatrician visits with ENT specialist.

Neurologist visits on Friday.

P is neither Orthopaedics nor Radiologist.

What is the specialty of P?

A-ENT

B-Ophthalmologist

C-Neurologist

D-Paediatrics

E-Radiologist

Q44

Study the following information and answer the question that follows:

Seven specialist doctors, P, Q, R, S, T, U and V, visit a polyclinic on any one of the four days in a week, i.e. on Tuesday, Wednesday, Friday or Saturday.

At least one doctor but not more than two doctors visit the polyclinic on each of these days. Each of them is a specialist in different fields - ENT, Orthopaedics, Paediatrics, Neurology, ophthalmology, Radiology and Oncology.

S visits on Friday with Radiologist, Paediatrician does not visit on Saturday nor with T and V.

Oncologist U visits alone on Tuesday.

Q visits on Wednesday and he is not a Paediatrician

R visits on Wednesday.

V is not a Radiologist.



Paediatrician visits with ENT specialist.

Neurologist visits on Friday.

P is neither Orthopaedics nor Radiologist.

On which day of the week does T visit?

- A-Wednesday
- B-Tuesday
- C-Friday
- D-Saturday
- E-Tuesday or Saturday

Q45

Which of the following alternatives will come next in the series?

21 25 18 29 33 18 ? ?

- A- 37,41
- B- 41,44
- C- 43,18
- D- 36,41
- E- 38,41

Q46

Analyze the question and decide whether the data given in the statements is sufficient to find an answer or not.

Question: A, B, C, D, and E scored different marks in different exams.
From statements given below determine who has secured the last position.

Statement I: D has scored fewer marks than C and E.

Statement II: B has scored more marks than A.

- A- Data in Statement 1 is sufficient alone to determine the answer
- B- Data in Statement 2 is sufficient alone to determine the answer
- C- Data in either of the statements is sufficient to determine the answer
- D- Data provided in both the statements together are not sufficient to determine the answer
- E- None of these

Q47



Analyze the question and decide whether the data given in the statements is sufficient to find an answer or not.

Question: Four players P, Q, R, and S are playing cards. Who is R's partner in the game?

The statement I: Q is to the right of P and left of S.

Statement II: S is opposite to the P.

- A- Data in Statement 1 is sufficient alone to determine the answer
- B- Data in Statement 2 is sufficient alone to determine the answer
- C- Data in either of the statements is sufficient to determine the answer
- D- Data provided in both the statements together are not sufficient to determine the answer
- E- None of these

Q48

Analyze the question properly and decide whether the data given in the statements is sufficient to find an answer or not.

Question: Parth was on which date in the month of February 2004?

The statement I: Parth's birthdate is a prime number.

Statement II: Parth was born on the even date of the month.

- A- Data in Statement 1 is sufficient alone to determine the answer
- B- Data in Statement 2 is sufficient alone to determine the answer
- C- Data in either of the statements is sufficient to determine the answer
- D- Data provided in both the statements together are not sufficient to determine the answer
- E- None of these

Q49

A cube is divided into 343 identical cubelets. Each cut is made parallel to some surface of the cube. But before doing that the cube is colored with green on one set of adjacent faces, red on the second and blue on the third set.

How many minimum cuts you have made?

- A-15
- B-18
- C-12
- D-7
- E-13

Q50



A cube is divided into 343 identical cubelets. Each cut is made parallel to some surface of the cube. But before doing that the cube is colored with green on one set of adjacent faces, red on the second and blue on the third set.

How many cubelets are coloured with exactly two colours?

- A-49
- B-55
- C-60
- D-25
- E-45

Q51

In the following question, two statements numbered 1 and 2 are given. Read both the statements and decide which of the following answer correctly depicts the relationship between these two statements:

Statements:

1. The prices of petrol and diesel in the domestic market have remained unchanged for the past few months.
2. The crude oil prices in the international market have gone up substantially in the last few months.

- A- Statement 1 is the cause and Statement 2 is its effect
- B- Statement 2 is the cause and Statement 1 is its effect
- C- Both the Statement 1 and 2 are independent causes
- D- Both the Statements 1 and 2 are effects of independent causes
- E- Both the Statements 1 and 2 are effects of some common cause

Q52

In the following question, two statements numbered 1 and 2 are given. Read both the statements and decide which of the following answer correctly depicts the relationship between these two statements:

Statements:

1. The government has recently fixed the fees for professional courses offered by the unaided institutions which are much lower than the fees charged last year.
2. The parents of the aspiring students launched a severe agitation last year protesting against the high fees charged by the unaided institutions.

- A- Statement 1 is the cause and Statement 2 is its effect
- B- Statement 2 is the cause and Statement 1 is its effect
- C- Both the Statement 1 and 2 are independent causes
- D- Both the Statements 1 and 2 are effects of independent causes
- E- Both the Statements 1 and 2 are effects of some common cause

Q53

In the following question, two statements numbered 1 and 2 are given. Read both the statements and decide which of the following answer correctly depicts the relationship between these two statements:

Statements:

1. The private medical colleges have increased the tuition fees in the current year by 200 per cent over the last year's fees to meet the expenses.
2. The Government medical colleges have not increased their fees in spite of price escalation.

- A- Statement 1 is the cause and Statement 2 is its effect
- B- Statement 2 is the cause and Statement 1 is its effect
- C- Both the Statement 1 and 2 are independent causes
- D- Both the Statements 1 and 2 are effects of independent causes
- E- Both the Statements 1 and 2 are effects of some common cause

Q54



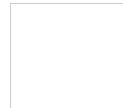
In the following question, two statements numbered 1 and 2 are given. Read both the statements and decide which of the following answer correctly depicts the relationship between these two statements:

Statements:

1. Most of the steel producing companies in the country have made considerable profit during the last financial year.
2. Many Asian countries have been importing huge quantities of steel from India.

- A- Statement 1 is the cause and Statement 2 is its effect
- B- Statement 2 is the cause and Statement 1 is its effect
- C- Both the Statement 1 and 2 are independent causes
- D- Both the Statements 1 and 2 are effects of independent causes
- E- Both the Statements 1 and 2 are effects of some common cause

Q55



Read the following passage and answer the question that follows:

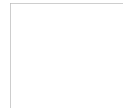
Sixty-four players seeded from seed 1 to seed 64 participated in a knock-out tennis tournament. Seed 1 is the highest seed and seed 64 is the lowest seed. The tournament will be played in six rounds i.e., first round, second round, third round, quarterfinals, semi-finals and final. In the first round, the player with the highest seed (i.e., 1) will play with the player with the lowest seed (i.e., 64), which is designated Match No.1.

Similarly, the player with the second-highest seed (i.e., 2) would play with the player with the second-lowest seed (i.e., 63), which is designated Match No.2 and so on. In the second round, the winner of the Match No.1 of the first round will play with the winner of the Match No.32 of the first round and this match is designated Match No.1 of the second round.

Similarly, the winner of the Match No.2 of the first round will play with the winner of the Match No.31 of the first round and this match is designated Match No.2 of the second round and so on. In the similar pattern the subsequent rounds will be played.

If the player seeded 43 won the tournament, then which of the following players cannot be the runner-up?

- A-Player seeded 42
- B-Player seeded 44
- C-Player seeded 46
- D-Player seeded 45
- E-Player seeded 40



Read the following passage and answer the question that follows:

Sixty-four players seeded from seed 1 to seed 64 participated in a knock-out tennis tournament. Seed 1 is the highest seed and seed 64 is the lowest seed. The tournament will be played in six rounds i.e., first round, second round, third round, quarterfinals, semi-finals and final. In the first round, the player with the highest seed (i.e., 1) will play with the player with the lowest seed (i.e., 64), which is designated Match No.1.

Similarly, the player with the second-highest seed (i.e., 2) would play with the player with the second-lowest seed (i.e., 63), which is designated Match No.2 and so on. In the second round, the winner of the Match No.1 of the first round will play with the winner of the Match No.32 of the first round and this match is designated Match No.1 of the second round.

Similarly, the winner of the Match No.2 of the first round will play with the winner of the Match No.31 of the first round and this match is designated Match No.2 of the second round and so on. In the similar pattern the subsequent rounds will be played.

Who could be the lowest seeded player facing the player seeded 12 in the finals?

- A-57
- B-63
- C-62
- D-59
- E-65

Q57

Read the following passage and answer the question that follows:

Sixty-four players seeded from seed 1 to seed 64 participated in a knock-out tennis tournament. Seed 1 is the highest seed and seed 64 is the lowest seed. The tournament will be played in six rounds i.e., first round, second round, third round, quarterfinals, semi-finals and final. In the first round, the player with the highest seed (i.e., 1) will play with the player with the lowest seed (i.e., 64), which is designated Match No.1.

Similarly, the player with the second-highest seed (i.e., 2) would play with the player with the second-lowest seed (i.e., 63), which is designated Match No.2 and so on. In the second round, the winner of the Match No.1 of the first round will play with the winner of the Match No.32 of the first round and this match is designated Match No.1 of the second round.

Similarly, the winner of the Match No.2 of the first round will play with the winner of the Match No.31 of the first round and this match is designated Match No.2 of the second round and so on. In the similar pattern the subsequent rounds will be played.

If one of the matches was between the players seeded 23 and 46, then one of the matches in the tournament can be between players seeded

- A-9 and 13
- B-6 and 18
- C-5 and 51
- D-20 and 35
- E-17 and 15

Q58

In the following question find out the alternative which will replace the question mark.

BFNM : EIQP :: RBGJ : ?

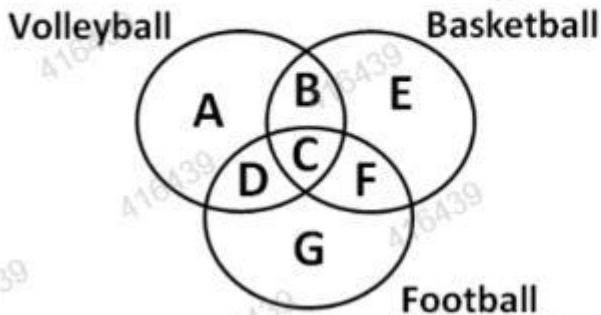
- A-GHJK
- B-WXYZ
- C-UEJM
- D-ABCD

E-PQRS



Q59

Which of the region in the following figure represents students who play Volleyball or Football or both but not Basketball?



- A- A+G
- B- A+D+G
- C- C+D
- D- A+C+D
- E- A+G+D

Q60

In a row of 26 girls, when Sakshi shifted four places towards the left, she become 10th from the left end. What was her earlier position from the right end of the row?

- A-10th
- B-14th
- C-15th
- D-12th
- E-13th

Q61

Read the following information carefully and answer the question that follows:

A + B means A is the son of B; A – B means A is the wife of B; A x B means A is the brother of B; A / B means A is the mother of B and A = B means A is the sister of B.

What does $P + R - Q$ mean?

- A-Q is the father of P
- B-Q is the son of P
- C- Q is the uncle of P

D- Q is the brother of P
E- Q is the nephew of P



Q62

Read the following information carefully and answer the question that follows:

A + B means A is the son of B; A – B means A is the wife of B; A x B means A is the brother of B; A / B means A is the mother of B and A = B means A is the sister of B.

What does P x R + Q mean?

- A-P is the sister of Q
- B-P is the father of Q
- C-P is the uncle of Q
- D-P is the nephew of Q
- E-P is the son of Q

Q63

Read the following information carefully and answer the question that follows:

A + B means A is the son of B; A – B means A is the wife of B; A x B means A is the brother of B; A / B means A is the mother of B and A = B means A is the sister of B.

What does P = R + Q mean?

- A-P is the daughter of Q
- B-P is the wife of Q
- C-P is the niece of Q
- D-P is the sister of Q
- E-P is the aunt of Q

Q64

Mohan said to Nitin, "That boy playing football is the younger of the two brothers of the daughter of my father's wife." How is the boy playing football related to Mohan?

- A-Son
- B-Brother
- C-Father

D-Nephew
E-Cousin



Q65

Veena who is the sister-in-law of Ashok, is the daughter-in-law of Kalyani. Dheeraj is the father of Sudeep who is the only brother of Ashok. How Kalyani is related to Ashok?

A-Sister-in-law
B-Sister
C-Mother
D-Daughter
E-Mother-in-law

Q66

A man is facing East. He turns 120° in the clockwise direction and then 210° in the anti-clock wise direction. Which direction is he facing now?

A-North
B-West
C-North-East
D-North-West
E-South-East

Q67

Samesh starts walking from a point a walks 12 kms towards north. From there he turns right and walks 4 km, then he again turns right and walks 9 km. How far and in what direction is he from his starting point?

A-10 km North
B-10 km North-East
C-5 km North
D-5 km North-East
E-6 km North

Q68

A fires 5 shots to B's 3 but A kills only once in 3 shots while B kills once in 2 shots. When B has missed 27 times, A has killed how many?

A-20
B-30
C-40
D-35
E-25



Q69

A Printer numbers the pages of a book starting with 1 and uses 3189 digits in all. How many pages does the book have?

- A-1000
- B-1074
- C-1025
- D-1080
- E-1098

Q70

Sapna travels 10 kms. to the north, turns left travels 4 km and then again turns right and cover another 5 km. And then turns right and travels another 4 km. How far Sapna from the starting point and in which direction she is from the starting point?

- A-15 km South
- B-10 km North
- C-15 km East
- D-15 km North
- E-10 km East

Q71



A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement:

Input: before its court seeking permission to carry out a test.

Step I: its before court seeking permission to carry out a test.

Step II: its before carryout court seeking permission to a test.

Step III: its before carryout court permission seeking to a test.

Step IV: a its before carryout court permission seeking to test.

Step V: a its before carryout court permission seeking test to.

As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

Which step is the last step for the given input?

Input: in this volume provides the

- A-1
- B-2
- C-3
- D-4
- E-5

Q72

Study the following information to answer the question that follows:

A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement:

Input: before its court seeking permission to carry out a test.

Step I: its before court seeking permission to carry out a test.

Step II: its before carryout court seeking permission to a test.

Step III: its before carryout court permission seeking to a test.

Step IV: a its before carryout court permission seeking to test.

Step V: a its before carryout court permission seeking test to.

As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

If the Input is: 'these many other clinicians and', which of the following steps would be 'and clinicians many these others'?

- A-1
- B-2
- C-3
- D-4
- E-5

Q73



Study the following information to answer the question that follows:

A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement:

Input: before its court seeking permission to carry out a test.

Step I: its before court seeking permission to carry out a test.

Step II: its before carryout court seeking permission to a test.

Step III: its before carryout court permission seeking to a test.

Step IV: a its before carryout court permission seeking to test.

Step V: a its before carryout court permission seeking test to.

As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

Step III of a certain line of words is like this – in play role schooling vital, which of the following will definitely be the input?

- A- in play schooling vital role
- B- schooling play vital role in
- C- in schooling play vital role
- D- in schooling play role vital
- E- in schooling vital role play



Study the following information to answer the question that follows:

A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement:

Input: before its court seeking permission to carry out a test.

Step I: its before court seeking permission to carry out a test.

Step II: its before carryout court seeking permission to a test.

Step III: its before carryout court permission seeking to a test.

Step IV: a its before carryout court permission seeking to test.

Step V: a its before carryout court permission seeking test to.

As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

Which of the following will be Step II for an input – modern man is becoming a chronic victim?

A- a becoming modern man is chronic victim

- B- a becoming chronic modern man is victim
- C- a modern man is becoming chronic victim
- D- a becoming chronic is man modern victim
- E- a modern becoming chronic is man victim

Q75

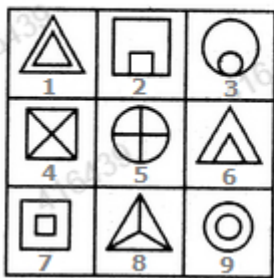
How many pairs of digits in the number **95137248** have the same number of digits between them as they would if the digits were arranged in ascending order?

- A-4
- B-2
- C-3
- D-5
- E-6

Q76



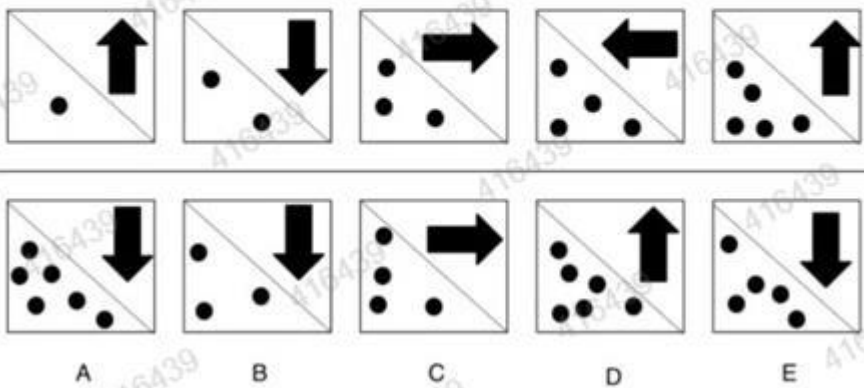
Group the given figures into three classes using each figure only once



- A- 1,7,9 2,3,6 4,5,8
- B- 1,2,9 3,4,6 5,7,8
- C- 1,6,8 2,4,7 3,5,9
- D- 1,7,8 2,9,3 6,4,5
- E- 1,2,9 3,6,7 4,5,8

Q77

Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?

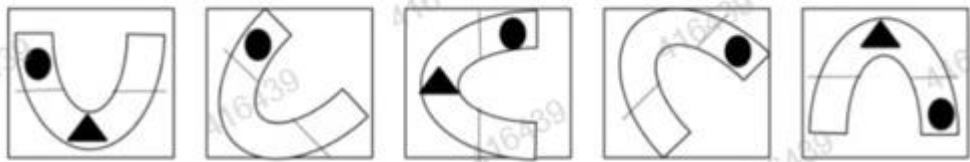


- A-A
- B-B
- C-C
- D-D
- E-E

Q78



Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?



A

B

C

D

E

A-A

B-B

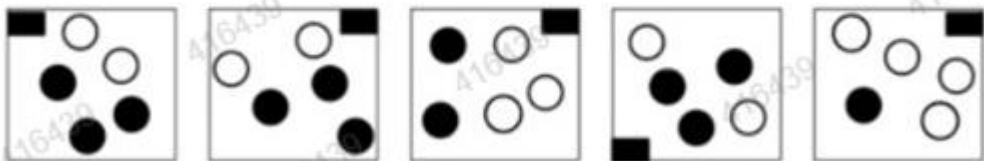
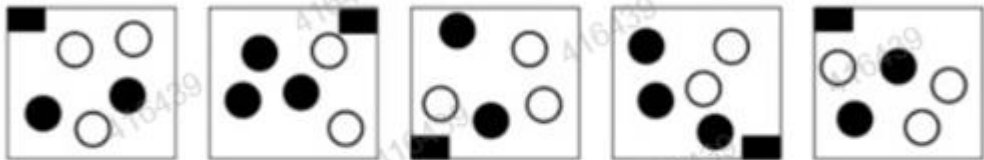
C-C

D-D

E-E

Q79

Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?



A

B

C

D

E

A-A

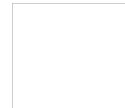
B-B

C-C

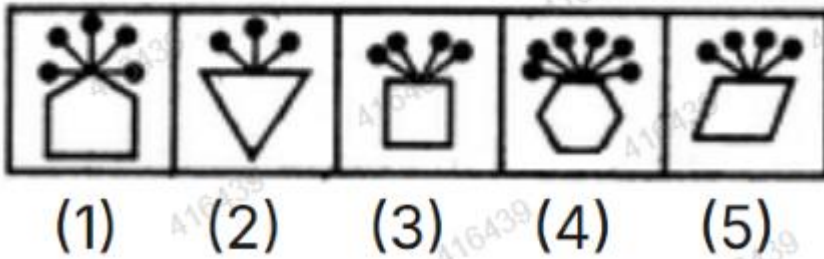
D-D

E-E

Q80



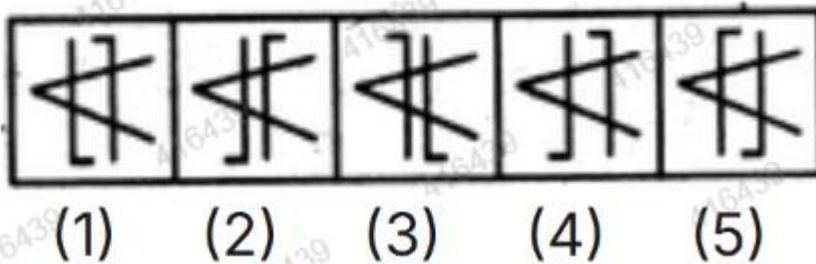
Choose the figure which is different from the rest.



- A-1
- B-2
- C-3
- D-4
- E-5

Q81

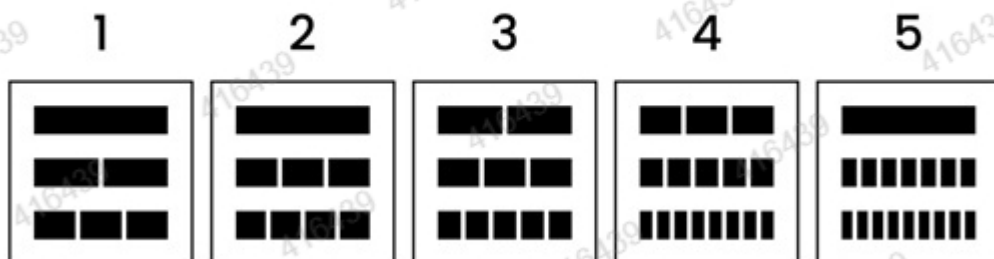
Choose the figure which is different from the rest.



- A-1
- B-2
- C-3
- D-4
- E-5

Q82

Select an item that does not belong to the group



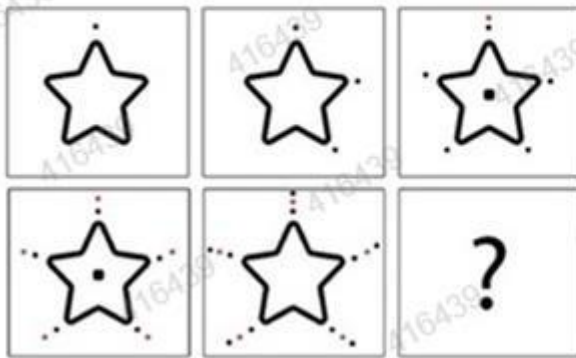
- A-1
- B-2

C-3
D-4
E-5



Q83

Choosing from options A, B, C, D, E, which figure completes the series below?



A

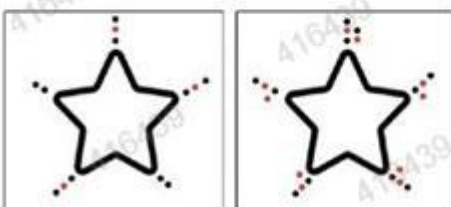
B

C



D

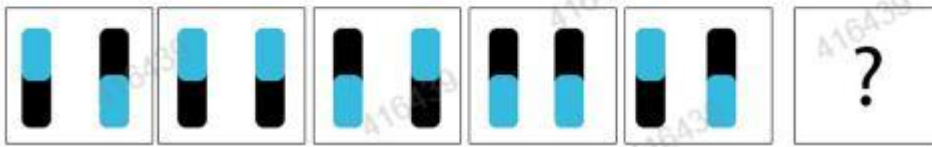
E



A-A
B-B
C-C
D-D
E-E

Q84

Choosing from options A, B, C, D, E, which figure will come next in the series given in the first row from left to right?



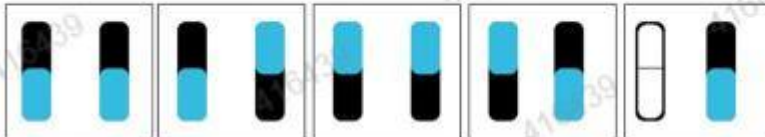
A

B

C

D

E



A-A

B-B

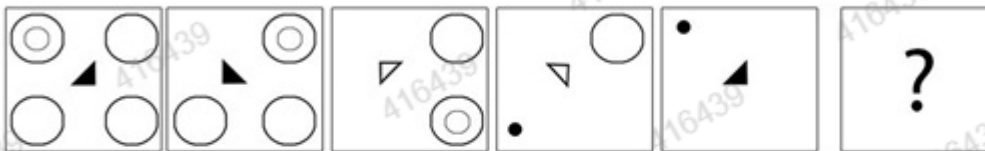
C-C

D-D

E-E

Q85

Choosing from options A, B, C, D, E, which figure completes the series below?



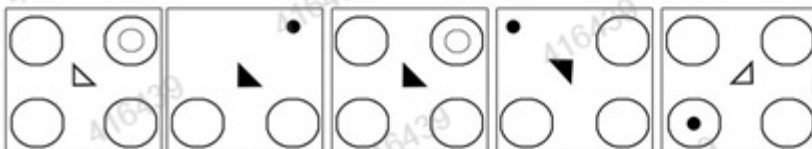
A

B

C

D

E



A-A

B-B

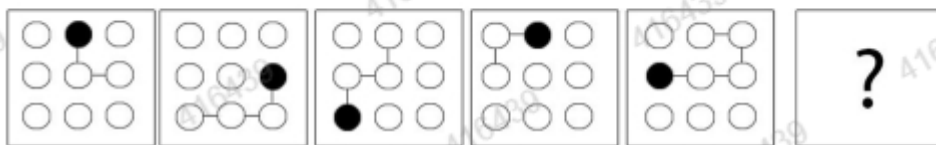
C-C

D-D

E-E

Q86

Choosing from options A, B, C, D, E, which figure will come next in the series given in the first row from left to right?



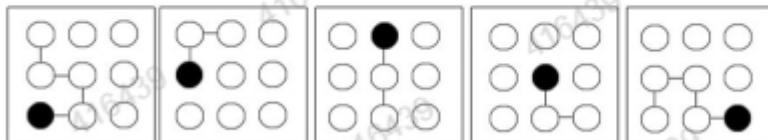
A

B

C

D

E



A-A

B-B

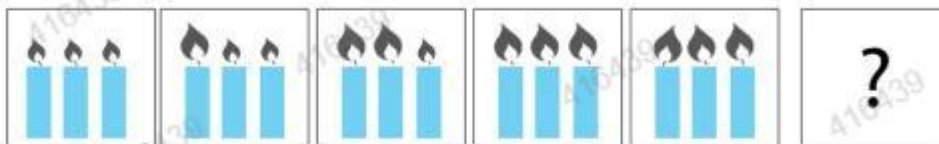
C-C

D-D

E-E

Q87

Choosing from options A, B, C, D, E, which figure will come next in the series given in the first row from left to right?



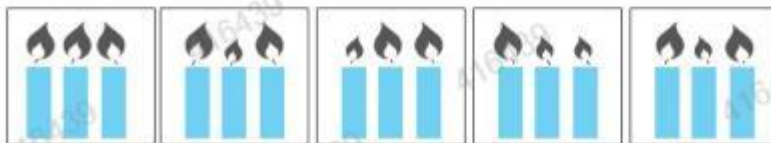
A

B

C

D

E



A-A

B-B

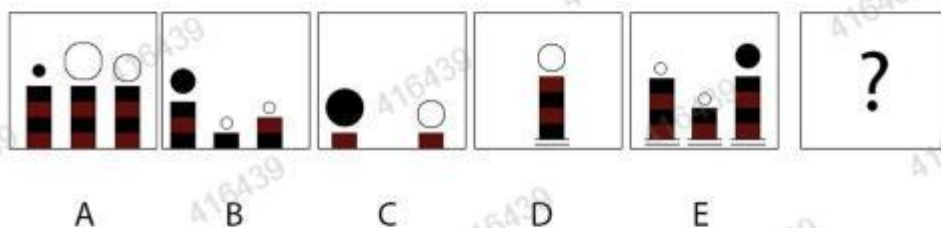
C-C

D-D

E-E

Q88

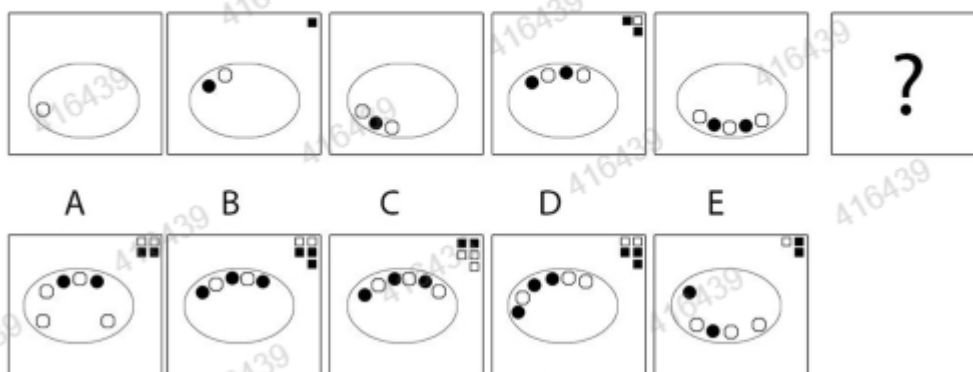
Choosing from options A, B, C, D, E, which figure will come next in the series given in the first row from left to right?



- A-A
- B-B
- C-C
- D-D
- E-E

Q89

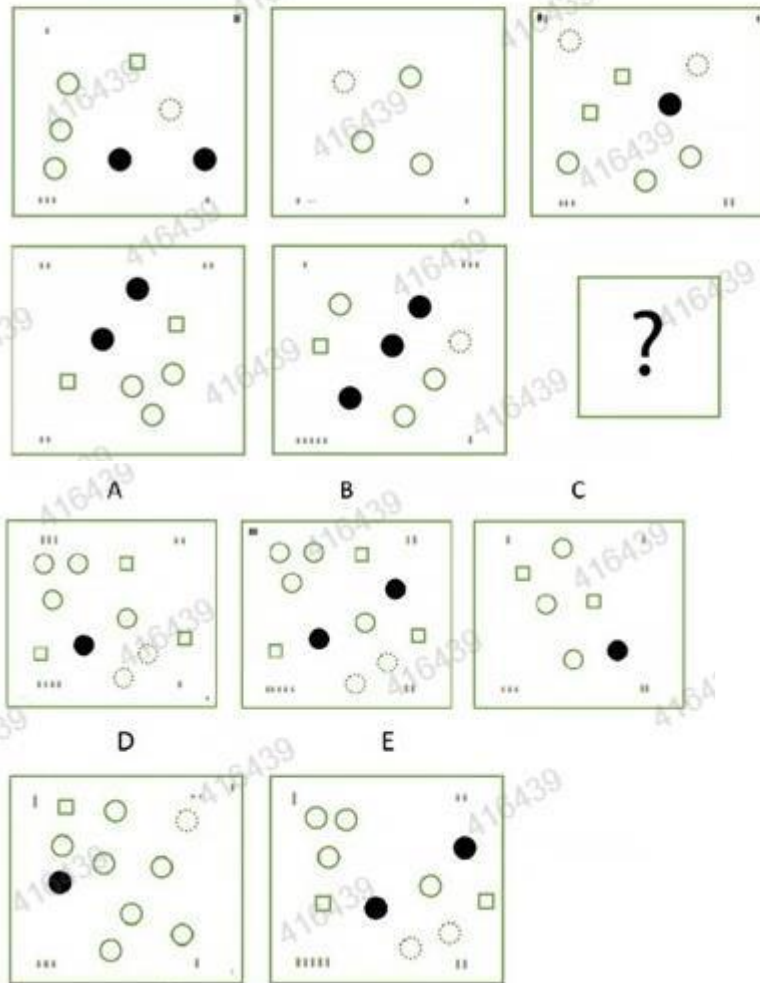
Choosing from options A, B, C, D, E, which figure will come next in the series given in the first row from left to right?



- A-A
- B-B
- C-C
- D-D
- E-E

Q90

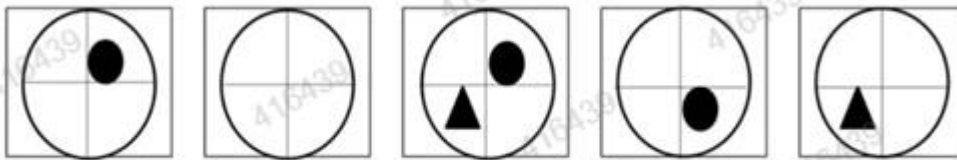
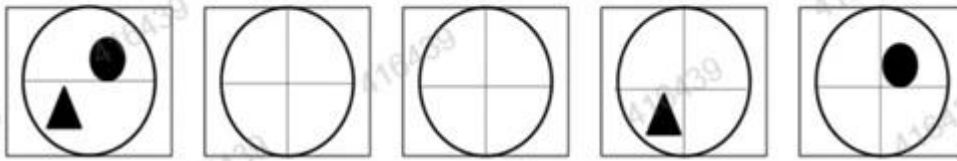
Choosing from options A, B, C, D, E, which figure completes the series below?



- A-A
- B-B
- C-C
- D-D
- E-E

Q91

Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?



A

B

C

D

E

A-A

B-B

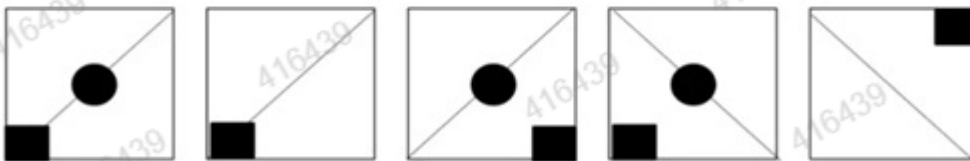
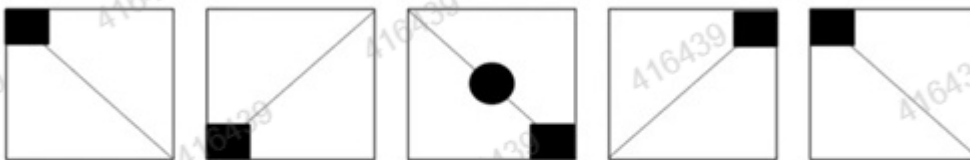
C-C

D-D

E-E

Q92

Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?



A

B

C

D

E

A-A

B-B

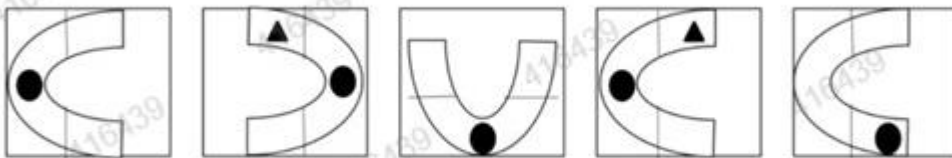
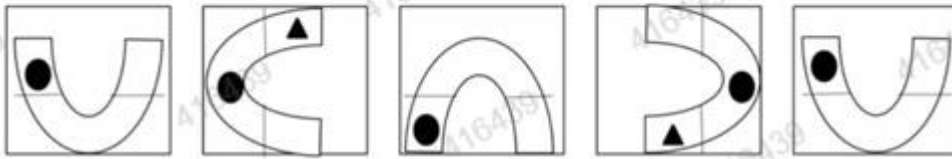
C-C

D-D

E-E

Q93

Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?



A

B

C

D

E

A-A

B-B

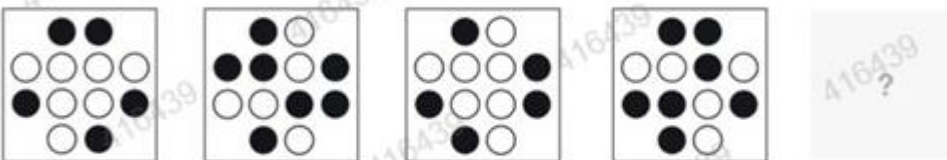
C-C

D-D

E-E

Q94

Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?



(A)

(B)

(C)

(D)

(E)



A-A

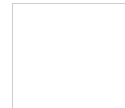
B-B

C-C

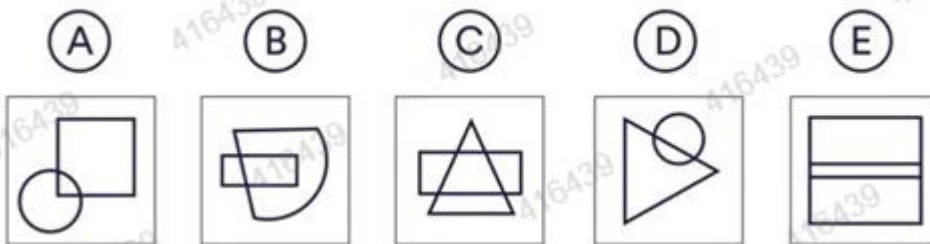
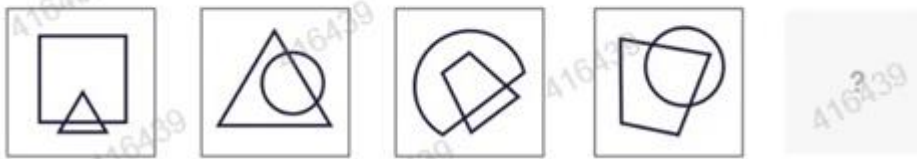
D-D

E-E

Q95



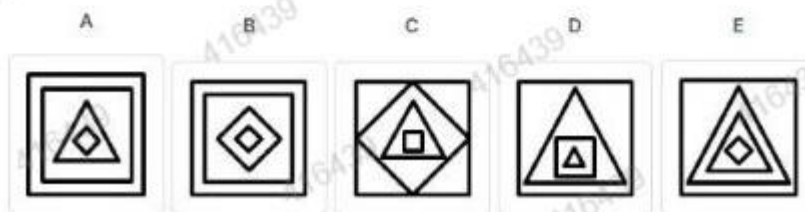
Which of the boxes from the given alternatives comes next in the sequence given in the top row from left to right?



- A-A
- B-B
- C-C
- D-D
- E-E

Q96

Choose the image from the given alternatives that completes the pattern in the top row (From left to right):



- A-A
- B-B
- C-C
- D-D
- E-E

Q97

Choose the image from the given alternatives that completes the pattern in the top row (From left to right):



A

B

C

D

E



A-A

B-B

C-C

D-D

E-E

Q98

Choose the odd one out:

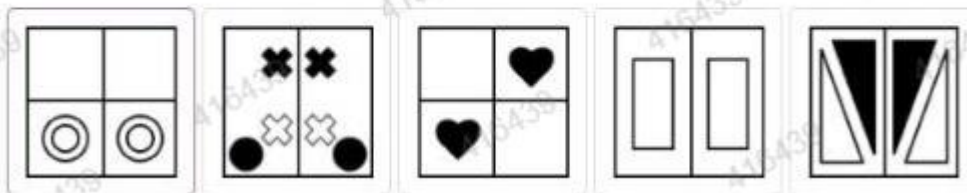
A

B

C

D

E



A-A

B-B

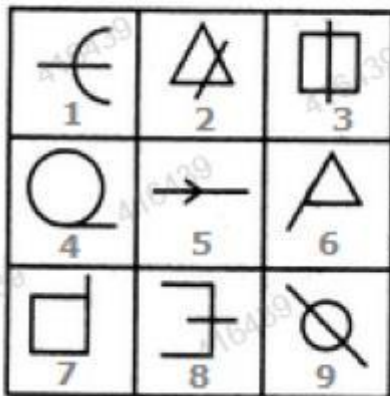
C-C

D-D

E-E

Q99

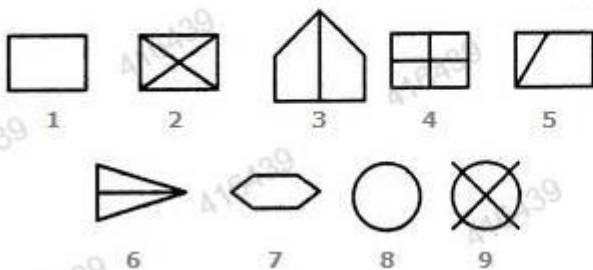
Group the given figures into three classes using each figure only once.



- A- 1,3,9 2,5,8 4,5,8
- B- 1,5,8 4,6,7 2,3,9
- C- 2,5,9 1,3,8 2,6,7
- D- 1,8,9 4,6,7 2,3,5
- E- 1,2,9 4,6,7 8,3,5

Q100

Group the given figures into three classes using each figure only once.



- A- 1,2,4 3,5,6 7,8,9
- B- 1,7,8 3,5,6 2,4,9
- C- 1,3,4 2,8,9 5,6,7
- D- 1,7,8 2,3,6 4,5,9
- E- 1,4,8 2,3,6 7,5,9

Q101

After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?

- A-50
- B-48
- C-53
- D-80

E-72



Q102

At a college football game, $\frac{4}{5}$ of the seats on the lower deck of the stadium were sold. If $\frac{1}{4}$ of all the seating in the stadium is located on the lower deck, and if $\frac{2}{3}$ of all the seats in the stadium were sold, what is the fraction of unsold seats on the lower deck to the total number of unsold seats?

A- $\frac{3}{20}$

B- $\frac{1}{20}$

C- $\frac{1}{2}$

D- $\frac{1}{5}$

E- $\frac{2}{5}$

Q103

Find the smallest number, which must be subtracted from 306452 to make it a perfect square.

A-1980

B-1748

C-1396

D-1558

E-1290

Q104

The least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is:

A-94

B-74

C-240

D-364

E-184

Q105

A shopkeeper has three kinds of sugar: 184 kg, 230 kg and 276 kg. He wants to store them in a minimum number of bags of equal size without mixing. Find the required number of bags.

A-18

B-15

C-17

D-20

E-16

Q106

The average mark of a class of n students is 64. If eight new students with an average mark of 73 join the class, the new average of the entire class will be a whole number. Find the number of students in the class (given that n lies between 25 and 60).

A-18

B-36

C-72

D-27

E-45

Q107

The age of the captain of a cricket team of 11 players is 25 years, the wicket keeper is 3 years older than the captain. If the ages of these two are excluded, the average age of the remaining players of the team becomes 1 year less than the average age of the whole team. What is the average age of the whole team?

A-25

B-22

C-26

D-28

E-24

Q108

The average number of printing errors in a book of 512 pages is 4. If the total number of printing errors on the first 302 pages is 1208, then what is the average number of printing errors on the remaining pages?

A-2

B-3

C-5

D-4

E-6

Q109

A woman sells to the first customer half of her stock of apples and half an apple, to the second customer she sells half her remaining stock and half an apple, and so on to the third and to the fourth customer. She finds that she now has 15 apples left. How many apples did she have before she started selling them?

A-250

B-255
C-254
D-270
E-350

Q110

A boy has enough money to buy 20 exercise books. If each book cost 25 paise less, he could buy two more books and still have 70 paise left. How much money did he have originally?

A-35
B-42
C-48
D-52
E-50

Q111

The digit at unit place of a two-digit number is increased by 100% and the digit at ten places of the same number is increased by 50%. The new number thus formed is 19 more than the original number. What is the original number?

A-22
B-25
C-27
D-29
E-21

Q112

The price of sugar was increased by 32%. A family reduces its consumption so that the expenditure on sugar is up by 10% only. If the total consumption of sugar before the price rises was 10 kg per month, then the consumption of sugar per month at present (in kg) is:

A-8.5
B-9
C-8.33
D-8.6
E-8.2

Q113

In a housing society, 30% of residents are men over the age of 18 and 40% are women over the age of 18. If there are 24 children living in the housing society, then how many total residents live in the society?

- A-75
- B-80
- C-72
- D-100
- E-32

Q114

The price of an item was increased by 10%. This reduced the monthly total sales by 20%. What is the overall effect on the value of monthly sales?

- A-12% decrease
- B-10% increase
- C-8% decrease
- D-10% decrease
- E-13% decrease

Q115

Two musical instruments were purchased for Rs. 8,000. The first was sold at a profit of 40% and the second at a loss of 40%. If the sale price was the same in both cases, then the cost price of the cheaper instrument (in Rs.) was:

- A-2500
- B-2400
- C-2200
- D-2000
- E-2300

Q116

Five kg of butter was bought by a shopkeeper for ₹300. One kg becomes unsaleable. He sells the remaining in such a way that, on the whole, he incurs a loss of 10%. At what price (in Rs. per kg) was the butter sold?

- A-67.50
- B-52.5
- C-72.5
- D-60
- E-55

Q117

A man sold two steel chairs for Rs. 500 each. On one, he gains 20% and on the other, he loses 12%. How much does he gain or lose in the whole transaction?

- A-1.5% gain
- B-1.5% loss
- C-2% gain

D-2% loss

E-1% loss



Q118

A manufacturer sells a pair of glasses to a wholesale dealer at a profit of 18%. The wholesaler sells the same to a retailer at a profit of 20%. The retailer in turn sells them to a customer for Rs.30.09, thereby earning a profit of 25%. What is the cost price of the pair of glasses to the manufacturer?

A-15

B-20

C-17

D-18

E-16

Q119

A rectangular paper, when folded into two congruent parts has a perimeter of 34 cm for each part folded along one set of sides and the same is 38 cm when folded along the other set of sides. What is the area of the paper (in sq. cms)?

A-130

B-140

C-150

D-160

E-145

Q120

Three containers have their volumes in the ratio 3 : 4 : 5. The containers are filled with a mixture of milk and water. The containers have milk and water in the ratio 4 : 1, 3 : 1 and 5 : 2 respectively. The contents of these three mixtures are poured into a fourth container. The ratio of milk and water in the fourth container is?

A-4:1

B-157:53

C-160:53

D-151:48

E-5:2

Q121

Two friends, P and Q started a business investing in the ratio 5 : 6. R joined them after six months, investing an amount equal to that of Q's. At the end of the year, 20% profit was earned which was equal to Rs. 98000. What was the amount invested by R?

- A-190000
- B-200000
- C-210000
- D-250000
- E-180000

Q122

A can contains a mixture of two liquids, 'A' and 'B' in the ratio 7 : 5. When 9 liters of mixture are drawn off and the can is filled with 'B', the ratio of 'A' and 'B' becomes 7 : 9 liters. How many liters of Liquid 'A' was initially contained by the can?

- A-15.25
- B-18
- C-21
- D-17.5
- E-16.75

Q123

The ratio between the length and the breadth of a rectangular park is 3 : 2. If a man cycling along the boundary of the park at the speed of 12 km/hr completes one round in 8 minutes, then what is the area of the park (in sq. m)?

- A-152600
- B-154600
- C-153600
- D-152500
- E-153500



Step 2: Use perimeter formula

Let:

- Length = $3x$
- Breadth = $2x$

Then:

$$\text{Perimeter} = 2(l + b) = 2(3x + 2x) = 10x$$

Set this equal to 1600:

$$10x = 1600 \Rightarrow x = 160$$

So:

- Length = $3x = 480$ m
 - Breadth = $2x = 320$ m
-

Step 3: Find the area

$$\text{Area} = \text{Length} \times \text{Breadth} = 480 \times 320 = \boxed{153600 \text{ m}^2}$$

Final Answer: The area of the park is 153600 square meters.

Q124

A, B and C start a business by investing Rs.2000, Rs.3000 and Rs.4000 respectively. But B increases his investment to Rs.4000 after 4 months and C withdraws Rs.1000 at the end of 9 months. What is A's share out of a total profit of Rs. 8475 earned in a year?

- A-1600
- B-1800
- C-1500
- D-1400
- E-1200

Q125



In a triangle ABC, points P, Q and R are the mid-points of the sides AB, BC and CA respectively. If the area of the triangle ABC is 20 sq. units, what is the area of the triangle PQR?

- A-10 Sq units
- B- 12 Sq units
- C- 5 Sq units
- D- 4 Sq units
- E- 15 Sq units

Q126

A train 125 m long passes a man, running at 5 km/hr in the same direction in which the train is going, in 10 seconds. The speed of the train is:

- A-45km/hrs.
- B-50km/hrs.
- C-54km/hrs.
- D-53km/hrs.
- E-55km/hrs.

Q127

The length of the bridge, which a train 130 metres long and travelling at 45 km/hr can cross in 30 seconds, is:

- A-200m
- B-230m
- C-245m
- D-250m
- E-220m

Q128

Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

- A-1:3
- B-3:2
- C-3:4
- D-1:4
- E-3:5

Q129

A train overtakes two persons walking along a railway track. The first walks at 4.5 km/hr. The other one walks at 5.4 km/hr. The train needs 8.4 and 8.5 seconds respectively to overtake them. What is the speed of the train if both the persons are walking in the same direction as the train?

- A-66km/hrs.
- B-76km/hrs.
- C-71km/hrs.
- D-81km/hrs.
- E-80km/hrs.

Q130

A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

- A-12
- B-15
- C-16
- D-18
- E-13

Q131

If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, the time taken by 15 men and 20 boys in doing the same type of work will be:

- A-4
- B-5
- C-6
- D-7
- E-3

Q132

A machine P can print one lakh books in 8 hours, machine Q can print the same number of books in 10 hours while machine R can print them in 12 hours. All the machines are started at 9 A.M. while machine P is closed at 11 A.M. and the remaining two machines complete work. Approximately at what time will the work (to print one lakh books) be finished ?

- A-11:30 Am
- B-12:30 Pm
- C-11:00 Am
- D-1:00 Pm
- E-1:30 Pm

Q133



A and B can do a work in 8 days, B and C can do the same work in 12 days. A, B and C together can finish it in 6 days. A and C together will do it in :

- A-4
- B-6
- C-8
- D-12
- E-10

Q134

The sum of ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child?

- A-4
- B-8
- C-10
- D-6
- E-5

Q135

Present ages of Sameer and Anand are in the ratio of 5 : 4 respectively. Three years hence, the ratio of their ages will become 11 : 9 respectively. What is Anand's present age in years?

- A-24
- B-27
- C-26
- D-29
- E-30

Q136

The present ages of three persons in proportions 4 : 7 : 9. Eight years ago, the sum of their ages was 56. Find their present ages (in years).

- A- 8,20,28 respectively
- B-16,28,36 respectively
- C-20,35,45 respectively
- D-18,32,49 respectively
- E-8,30,45 respectively

Q137

From a group of 7 men and 6 women, five persons are to be selected form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

- A-700
- B-695
- C-756
- D-823
- E-633
- Q138

In how many ways can the letters of the word 'LEADER' be arranged?

- A-360
- B-240
- C-180
- D-210
- E-270

Q139

How many 4-letter words with or without meaning, can be formed out of the letters of the word, 'LOGARITHMS', if repetition of letters is not allowed?

- A-6000
- B-5075
- C-5040
- D-3500
- E-4090

Q140

The cube root of 0.000216 is:

- A-0.6
- B-0.06
- C-0.006
- D-0.16
- E-0.016

Q141

In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. The duration of the flight is:

- A-1 hrs.
- B-2 hrs.
- C-3 hrs.

D-4 hrs.
E-2.5 hrs.



Q142

Two ships are sailing in the sea on the two sides of a lighthouse. The angle of elevation of the top of the lighthouse is observed from the ships are 30° and 45° respectively. If the lighthouse is 100 m high, the distance between the two ships is:

- A-270m
- B-273m
- C-173m
- D-275m
- E-175m

Q143

A hollow iron pipe is 21 cm long and its external diameter is 8 cm. If the thickness of the pipe is 1 cm and iron weighs 8 g/cm^3 , then the weight of the pipe is:

- A-3.6kg
- B-3.69kg
- C-3.696kg
- D-3.96kg
- E-3.66kg

Q144

A boat running upstream takes 8 hours 48 minutes to cover a certain distance, while it takes 4 hours to cover the same distance running downstream. What is the ratio between the speed of the boat and speed of the water current respectively?

- A-2:1
- B-2:3
- C-8:3
- D-3:5
- E-5:8

Q145

A boat takes 90 minutes less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 mph, the speed of the stream is:

- A-2mph

- B-2.5mph
C-3mph
D-4mph
E-3.2mph
Q146



A man can row three-quarters of a kilometer against the stream in 11 minutes 15 seconds and down the stream in 7 minutes 30 seconds. What is the speed of the man (in km/hr) in still water?

- A-1
B-2
C-3
D-4
E-5

Q147

Pipe A can fill a cistern in 37 minutes 30 seconds and Pipe B fills it in 45 minutes individually. If both pipes are opened together and pipe B is turned off after sometime, it takes half an hour for the cistern to be filled. After how much time pipe B is turned off?

- A-8 mins
B-9 mins
C-10 mins
D-12 mins
E-6 mins

Q148

Two pipes A and B can fill a tank in 15 minutes and 20 minutes respectively. Both the pipes are opened together but after 4 minutes, pipe A is turned off. What is the total time required to fill the tank?

- A-10 min 20sec
B-12 min 20sec
C-14 min 40sec
D-14 min 20sec
E-10 min 40sec

Q149

The angle between the minute hand and the hour hand of a clock when the time is 4.20, is:

- A-3 degree
B-5 degree
C-8 degree
D-10 degree

E-12 degree



Q150

A and B take part in 100 m race. A runs at 5 kmph. A gives B a start of 8 m and still beats him by 8 seconds. The speed of B is:

- A-4.14kmph
- B-4.5kmph
- C-4.32kmph
- D-4.9kmph
- E-4.11kmph

Q151

The sentences given in each question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the given choices to construct a coherent paragraph.

- A. Branded disposable diapers are available at many supermarkets and drug stores.
- B. If one supermarket sets a higher price for a diaper, customers may buy that brand elsewhere.
- C. By contrast, the demand for private-label products may be less price sensitive since it is available only at a corresponding supermarket chain.
- D. So the demand for branded diapers at any particular store may be quite price sensitive.
- E. For instance, only SavOn Drugs stores sell SavOn Drugs diapers.
- F. Then stores should set a higher incremental margin percentage for private label diapers

- A-ABCDEF
- B-ABCEDF
- C-ADBCEF
- D-AEDBCF
- E-AEDCFB

Q152

The sentences given in each question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the given choices to construct a coherent paragraph.

- A. Having a strategy is a matter of discipline.
- B. It involves the configuration of a tailored value chain that enables a company to offer unique value.
- C. It requires a strong focus on profitability and a willingness to make tough trade-offs in choosing what not to do.
- D. Strategy goes far beyond the pursuit of best practices.
- E. A company must stay the course even during times of upheaval, while constantly improving and extending its distinctive positioning.
- F. When a company's activities fit together as a self-reinforcing system, any competitor wishing to imitate a strategy must replicate the whole system.

- A-ACEDBF
- B-ACBDEF
- C-DCBEFA
- D-ABCEDF
- E-DABCEF

Q153

The sentences given in each question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the given choices to construct a coherent paragraph.

- A. As officials, their vision of a country shouldn't run too far beyond that of the local people with whom they have to deal.
- B. Ambassadors have to choose their words.



C. To say what they feel they have to say, they appear to be denying or ignoring part of what they know.

D. So, with ambassadors as with other expatriates in black Africa, there appears at a first meeting a kind of ambivalence.

E. They do a specialized job and it is necessary for them to live ceremonial lives.

A-BCEDA

B-BEDAC

C-BEADC

D-BCDEA

E-BACDE

Q154

The sentences given in each question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the given choices to construct a coherent paragraph.

A. "This face-off will continue for several months given the strong convictions on either side," says a senior functionary of the high-powered task force on drought.

B. During the past week-and-half, the Central Government has sought to deny some of the earlier apprehensions over the impact of drought.

C. The recent revival of the rains had led to the emergence of a line of divide between the two.

D. The state governments, on the other hand, allege that the Centre is downplaying the crisis only to evade its full responsibility of financial assistance that is required to alleviate the damage.

E. Shrill alarm about the economic impact of an inadequate monsoon had been sounded by the Centre as well as most of the states, in late July and early August.



- A-EBCDA
- B-DBACE
- C-BDCAE
- D-ECBDA
- E-ECBAD

Q5

The sentences given in each question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the given choices to construct a coherent paragraph.

- A. This fact was established in the 1730s by French survey expeditions to Equador near the Equator and Lapland in the Arctic, which found that around the middle of the earth the arc was about a kilometer shorter.
- B. One of the unsettled scientific questions in the late 18th century was that of exact nature of the shape of the earth.
- C. The length of one-degree arc would be less near the equatorial latitudes than at the poles.
- D. One way of doing that is to determine the length of the arc along a chosen longitude or meridian at one degree latitude separation.
- E. While it was generally known that the earth was not a sphere but an 'oblate spheroid', more curved at the equator and flatter at the poles, the question of 'how much more' was yet to be established.

- A-BECAD
- B-BEDCA
- C-EDACB
- D-EBDCA
- E-BEDAC

Q156

Fill the gaps in the passages below with the most appropriate word from the options given for each gap. The right words are the ones used by the author. Be guided by the author's overall style and meaning when you choose the answers.



Von Nuemann and Morgenstern assume a decision framework in which all options are thoroughly considered, each option being independent of the others, with a numerical value derived for the utility of each possible outcome (these outcomes reflecting, in turn, all possible combinations of choices). The decision is then made to maximize the expected utility.

... 1 ... such a model reflects major simplifications of the way divisions are made in the real world. Humans are not able to process information as quickly and effectively as the model assumes; they tend not to think ... 2 ... as easily as the model calls for; they often deal with a particular option without really assessing its ... 3 ... and when they do assess alternatives, they may be extremely nebulous about their criteria of evaluation.

In a large company, firing people is about as common as using a gun or a switch-blade to ... 4 ... an argument. As a result, most managers have little or no experience of firing people, and they find it emotionally traumatic; as result, they often delay the act interminably, much as an unhappy spouse will prolong a bad marriage. And when the firing is done, it's often done clumsily, with far worse side effects than are necessary. Do the world-class software organizations have a different way of firing people? No, but they do the deed swiftly, humanely, and professionally. The key point here is to view the fired employee as a 'failed product' and to ask how the process ... 5 ... such a phenomenon in the first place.

1

- A-Regrettably
- B-Firstly
- C-Obviously
- D-Apparently
- E-Important

Q157



Fill the gaps in the passages below with the most appropriate word from the options given for each gap. The right words are the ones used by the author. Be guided by the author's overall style and meaning when you choose the answers.

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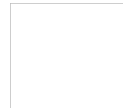
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2

- A-Quantitatively
- B-Systematically
- C-Scientifically
- D-Analytically
- E-Interestingly

Q158



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3

- A-Implication
- B-Disadvantages
- C-Utility
- D-Alternatives
- E-Assets

Q159



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... 1 ... such a model reflects major simplifications of the way divisions are made in the real world. Humans are not able to process information as quickly and effectively as the model assumes; they tend not to think ... 2 ... as easily as the model calls for; they often deal with a particular option without really assessing its ... 3 ... and when they do assess alternatives, they may be extremely nebulous about their criteria of evaluation.

In a large company, firing people is about as common as using a gun or a switch-blade to ... 4 ... an argument. As a result, most managers have little or no experience of firing people, and they find it emotionally traumatic; as result, they often delay the act interminably, much as an unhappy spouse will prolong a bad marriage. And when the firing is done, it's often done clumsily, with far worse side effects than are necessary. Do the world-class software organizations have a different way of firing people? No, but they do the deed swiftly, humanely, and professionally. The key point here is to view the fired employee as a 'failed product' and to ask how the process ... 5 ... such a phenomenon in the first place.

4

- A-Resolve
- B-Thwart
- C-Defeat
- D-Close
- E-Assume

Q160



Fill the gaps in the passages below with the most appropriate word from the options given for each gap. The right words are the ones used by the author. Be guided by the author's overall style and meaning when you choose the answers.

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5

- A-Derived
- B-Engineered
- C-Produced
- D-Allowed
- E-Delived

Q161

Select the correct synonym of the given word.

Taciturnity

- A-Calmness
- B-Strength
- C-Precious
- D-Courage
- E-Beauty

Q162

Select the correct synonym of the given word.

Attenuate

- A-Repent
- B-Make Thin
- C-Force
- D-Divide
- E-Rule

Q163

Select the correct synonym of the given word.

Haughty

- A-Cemetery
- B-Adamant
- C-Imperious
- D-Recover
- E-Deadly

Q164

Select the correct synonym of the given word.

Reprisal

- A-Indulgence
- B-Retribution
- C-Expanse
- D-Reluctance
- E-Cautious

Q165



Select the correct synonym of the given word.

Condescend

- A-Mystery
- B-Deride
- C-Joy
- D-Distain
- E-Comprehend

Q166

Select the correct antonym of the given word.

Axiom

- A-Scenic
- B-Flashy
- C-Absurdity
- D-Subtle
- E-Dark

Q167

Select the correct antonym of the given word.

Repudiate

- A-Emphasize
- B-Abandon
- C-Decline
- D-Modest
- E-Orient

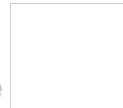
Q168

Select the correct antonym of the given word.

Remonstrate

- A-Dictate
- B-Neglect
- C-Endorse
- D-Strengthen
- E-Abandon

Q169



In the question given below, out of the given alternatives, choose the which can be substituted for the given word/sentence.

Extreme old age when a man behaves like a fool

- A-Nostalgia
- B-Serendipity
- C-Dotage
- D-Superannuation
- E-Ethnocity

Q170

In the question given below, out of the given alternatives, choose the one which can be substituted for the given word/sentence.

One who believes that all things and events in life are predetermined

- A-Extremist
- B-Fatalist
- C-Tyrant
- D-Puritan
- E-Egalitarian

Q171

Read each sentence to find out whether there is any grammatical error in it. The error, if any, will be in one part of the sentence. (Ignore the errors of punctuation, if any).

He fell from a running train and would have died if the villagers did not get him admitted to a nearby hospital immediately.

- A- He fell from a running train
- B- and would have died
- C- if the villagers did not get him admitted
- D- to a nearby hospital immediately.
- E- No error

Q172



Read each sentence to find out whether there is any grammatical error in it. The error, if any, will be in one part of the sentence.

He deserted the path of honor in order to satisfy his ambition and then went down his doom very quickly.

- A- He deserted the path of honor
- B- in order to
- C- satisfy his ambition
- D- and then went down his doom very quickly.
- E- No error

Q173

Read each sentence to find out whether there is any grammatical error in it. The error, if any, will be in one part of the sentence.

This is one of the most interesting book I have ever read.

- A- This is one of the
- B- most interesting book
- C- I have
- D- ever read.
- E- No error

Q174

Rearrange the following five sentences in a proper sequence to form a meaningful paragraph.

1. After examining him, the doctor smiled at him mischievously and took out a syringe.
2. Thinking that he was really sick, his father summoned the family doctor.
3. That day, Mintu wanted to take a day off from school.
4. Immediately, Mintu jumped up from his bed and swore that he was fine.
5. Therefore, he pretended to be sick and remained in bed.

A-35214

B-35241

C-15234

D-34215

E-35412

Q175

Rearrange the following five sentences in the proper sequence to form a meaningful paragraph.

1. Would you steal a software programme out of a retail shop?
2. The industry, on its part, has formed an organization to specially gather information and drag software pirates to the courts.
3. But more than the legality, there is always a different way of looking at piracy and that is in terms of morality.
4. The Government, on the other hand, has initiated National Enforcements Committees.
5. As far as the issue of tackling piracy is concerned, both the industry and the government have already started initiating action.

A-15243

B-53241

C-52413

D-52431

E-51243

Q176

A proverb/idiom is given below together with their meanings. Choose the correct meaning of the proverb/idiom.

To make clean breast of



- To gain prominence
- A-
- To praise oneself
- B-
- To confess without reserve
- C-
- To destroy before it blooms
- D-
- To appreciate someone
- E-

Q177

A proverb/idiom is given below together with their meanings. Choose the correct meaning of the proverb/idiom.

- To catch a tartar
- To trap wanted criminal with great difficulty
- A-
- To catch a dangerous person
- B-
- To meet with disaster
- C-
- To deal with a person who is more than one's match
- D-
- To meet an old friend
- E-

Q178

A proverb/idiom is given below together with their meanings. Choose the correct meaning of the proverb/idiom.

- To have an axe to grind
- To fail to arouse interest
- A-
- To have no result
- B-
- To work for both sides
- C-



A private end to serve

D-

To be unbiased

E-

Q179

A proverb/idiom is given below together with their meanings. Choose the correct meaning of the proverb/idiom.

To pick holes

to find some reason to quarrel

A-

to destroy something

B-

to criticise someone

C-

to cut some part of an item

D-

to find escape routes

E-

Q180

A proverb/idiom is given below together with their meanings. Choose the correct meaning of the proverb/idiom.

gift horse in the mouth

welcome a gift

A-

find fault with a gift

B-

be fastidious

C-

to be kind

D-

examine a gift carefully

E-

Q181

In modern times, Abraham Lincoln stands as the model of a compassionate statesman. He showed this quality not only in striving for the emancipation of American blacks but in the dignity with which he conducted the American Civil War. Lincoln did not fancy himself a liberator. He thought it would be better for all if emancipation was a gradual process spread over many years.

He proposed compensation for slave-owners in US bonds and grants for the rehabilitation of blacks— 'colonization' as he called it. But fate was to deem otherwise. The haste with which the South wanted to break away from the Union with the North, compelled him to move faster than he expected. Perhaps more than most men of his time, he had thought through the issue of slavery. "We must free the slaves", he said, "or be ourselves subdued". Before reading his first draft of the Proclamation of Emancipation, he told his colleagues, "In giving freedom to the slaves, we assure freedom to the free".

On September 22, 1862, Lincoln set his hand on the Proclamation of Emancipation declaring that on the first day of January 1863, all persons held as slaves within any state shall be then and forever free.

Lincoln's revulsion towards slavery left him without any moral indignation or passion against the slave – owners. The guilt of the slave – owners, he felt, should be shared by the whole country, the North and the South, for it seemed to him that everyone in the nation was an accomplice in perpetuating that system. To have whipped up any hatred against slave – owners would, to him, have been an act of malice.

"I shall do nothing in malice", he wrote, "What I deal with is too vast for malicious dealing".

As the Civil War was coming to a successful conclusion, a Northerner demanded of Lincoln, "Mr President, how are you going to treat the southerners when the war is over?". Lincoln replied, "As if they never went to war?"



When the news came of the victory of the north against the Confederate forces, someone suggested that the head of the confederation Administration, Jefferson Davies, really ought to be hanged. "Judge not, that yet be not judged". Lincoln replied as to the demand for the prosecution of rebels. He said, "We must extinguish our resentments if we expect harmony and union". This was his last recorded utterance.

What came in Lincoln's way of carrying out emancipation as a gradual process?

- A- His proposition to give a grant for the rehabilitation of slaves
 - B- His own over-enthusiasm to complete the process fast
 - C- The inadequate compensation was given to slave – owners
 - D- The haste of the South to break away from the union with the North
 - E- His ownership mindset
- Solution
- The haste of the South to break away from the union with the North

Q182

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The sentence 'In giving freedom to the free' means

- A- if we give freedom to the slaves, they will serve us better
- B- if we do not give freedom to the slaves, they will free themselves
- C- by freeing slaves, we are honouring the concept of freedom
- D- by freeing slaves, we are safeguarding our own interests
- E- by freeing slaves, we are helping them

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The term 'colonization' as used in the passage means

handing over slaves to the slave-owners.

A-

efforts made by American blacks to free themselves

B-

making separate dwelling arrangements for the slave – owners.

C-

Rehabilitation arrangements made for slave-owners

D-

colony of slaves

E-

Q184



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Which of the following makes Abraham Lincoln a compassionate statesman?

- A- His hesitation in striving for the emancipation of American blacks
- B- His efforts to force the American blacks from slavery
- C- His care and compassion towards people
- D- His indifference in conducting the American Civil War
- E- His way of gradual emancipation

Q185

In the question given below, a part of the sentence is italicized and underlined. Choose the correct alternative to replace the italicized part in order to improve the sentence.

The workers are hell bent at getting what is due to them

- A- hell bent on getting
- B- hell bent for getting
- C- hell bent upon getting
- D- hell bent in getting
- E- No improvement

Q186

In the question given below, a part of the sentence is italicized and underlined. Choose the correct alternative to replace the italicized part in order to improve the sentence.

If the room had been brighter, I would have been able to read for a while before bedtime.

- A- If the room was brighter
- B- If the room are brighter
- C- Had the room been brighter
- D- If the room had been brighter
- E- No improvement

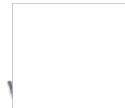
Q187

In the question below, the sentence is given in Direct speech. From the given alternatives, choose the one which best expresses the given sentence in Indirect speech.

He said to his father, "Please increase my pocket-money."

- A- He told his father, "Please increase the pocket-money"
- B- He pleaded his father to please increase my pocket money.
- C- He requested his father to increase his pocket-money.
- D- He asked his father to increase his pocket-money.
- E- He asked his father if he could increase his pocket-money.

Q188



The question consists of two words which have a certain relationship with each other, followed by alternative pairs of related words. Select the pair which has the same relationship.

INDIGENT : WEALTHY

- A- angry:rich
- B- native:affluent
- C- gauche:disgraceful
- D- scholarly:erudite
- E- bushel:corn

Q189

The question consists of two words which have a certain relationship with each other, followed by alternative pairs of related words. Select the pair which has the same relationship.

CORPOREAL : SPIRITUAL

- A- mesa:plateau
- B- moron:savant
- C- foreigner:immigrant
- D- pedagogue:teacher
- E- colours:pallet

Q190

The sentence is divided into four parts. Choose the part that is/are grammatically incorrect.

Those who want to do good are/ neither selfish nor in a hurry because/ they know what it requires a long/ time to impregnate people with good.

- A- Those who want to do good are

-
- neither selfish nor in a hurry because
- B- they know what it requires a long
- C- time to impregnate people with good.
- D- No error
- E-

Q191

Read the following passage carefully and answer the question that follows.

- (1) One of the most recent developments in tools is the robot. The latest robots can be taught to perform complex tasks that require an almost human-like ability to process information and make decisions. The word robot first appeared in a 1921 stage play by Czech writer Karel Capek. In the play, a man makes a machine that can think, which he calls a robot and which ends up killing its owner. In the 1940s, the American science fiction writer Isaac Asimov wrote a series of stories about robots and invented the term robotics, the science of robots. Meanwhile, in the real world, the first robots were developed by an engineer, Joseph F. Engel Berger, and an inventor, George C. Devol. Together they started a manufacturing company that produced the first real robot in 1961. Robots of this type were installed at automobile plants and proved to be a success. They worked reliably and saved money for companies.
- (2) These industrial robots were nothing like the terrifying creatures that can often be seen in science fiction films. Since they were made of metal, they could perform certain jobs that were difficult or dangerous for humans, particularly jobs that involve high heat. And since robots were tireless and never got hungry, sleepy, or distracted, they were useful for tasks that would be tiring or boring for humans. Industrial robots have been improved over the years, and today they are used in many factories around the world.



(3) Though the use of robots has meant the loss of some jobs, at the same time other jobs have been created in the design, development, and production of the robots. Outside of industry, robots have also been developed and put to use by governments and scientists in situations where humans might be in danger. For example, they can be sent in to investigate an un-exploded bomb or an accident at a nuclear power plant. Researchers also use robots to collect samples of hot rocks or gases on active volcanoes. In space exploration, robots have performed many key tasks where humans could not be present, such as on the surface of Mars. In 2004, two robotic Rovers—small six-wheeled computerized cars—were sent to Mars. The Rovers had lasers that functioned as eyes and software that was designed to help them travel around holes or rocks. While on Mars, they performed many scientific experiments and took thousands of photographs, sending all the results back to Earth.

(4) Inventors have been more successful with robots that can help with housework. "Roomba" is made of lightweight plastic, it is low and round, with wheels. Before it begins cleaning a room, Roomba first moves all around it to "learn" about the shape of the room and the furniture in it. Then it starts vacuuming, avoiding the walls and furniture.

(5) What is the future for robots? One likely use of robots was demonstrated in a race for robot-controlled cars held in 2005 by the United States Defense Department. The winner of this race across the Nevada desert—and the winner of \$2 million—was a car model called Stanley, designed by robot specialists at a renowned university. Stanley arrived at the finish line 170 miles (273.5 km) away, after traveling over the rough land in a little less than seven hours—without a human behind the steering wheel. This was possible because its computer "brain" was able to deal with the various tasks required along the trip.



(6) Robots have come a long way. Once, story writers and movie-makers imagined them as dangerous humans, like machines that could take over the Earth. Now, robots help people at home, in factories, and in scientific research. In the next decade, many new uses will be found for robots, especially in cars. As one designer said, "A person's role in the car is changing. People will become more planners than drivers. Robots will do the rest".

Choose the statement that best expresses the overall thesis of the passage.

- The word robot was invented by a Czech writer in 1921.
- A- Some kinds of robots are used in industry and scientific research.
 - B- Robots can be used to perform many complex tasks these days.
 - C- In the future, people won't drive cars—robots will.
 - D- the use of robots has meant the loss of some jobs
 - E-

Read the following passage carefully and answer the question that follows.

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The first real robot was

- A- installed in an automobile plant
- B- a machine that could think like a person
- C- invented by a science fiction writer
- D- looked and behaved like a human
- E- invented in 1921

Q193



Read the following passage carefully and answer the question that follows.

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(4) Inventors have been more successful with robots that can help with housework. "Roomba" is made of lightweight plastic, it is low and round, with wheels. Before it begins cleaning a room, Roomba first moves all around it to "learn" about the shape of the room and the furniture in it. Then it starts vacuuming, avoiding the walls and furniture.

(5) What is the future for robots? One likely use of robots was demonstrated in a race for robot-controlled cars held in 2005 by the United States Defense Department. The winner of this race across the Nevada desert—and the winner of \$2 million—was a car model called Stanley, designed by robot specialists at a renowned university. Stanley arrived at the finish line 170 miles (273.5 km) away, after traveling over the rough land in a little less than seven hours—without a human behind the steering wheel. This was possible because its computer "brain" was able to deal with the various tasks required along the trip.



(6) Robots have come a long way. Once, story writers and movie-makers imagined them as dangerous humans, like machines that could take over the Earth. Now, robots help people at home, in factories, and in scientific research. In the next decade, many new uses will be found for robots, especially in cars. As one designer said, "A person's role in the car is changing. People will become more planners than drivers. Robots will do the rest".

What kind of jobs are done by Industrial robots?

They saved money for companies

A-

Industrial robots perform jobs that require high heat

B-

They perform all kind of jobs at workplace

C-

Industrial robots perform jobs that involve less humans

D-

Industrial robots perform jobs that require less strength

E-

Q194



Read the following passage carefully and answer the question that follows.

(1) One of the most recent developments in tools is the robot. The latest robots can be taught to perform complex tasks that require an almost human-like ability to process information and make decisions. The word robot first appeared in a 1921 stage play by Czech writer Karel Capek. In the play, a man makes a machine that can think, which he calls a robot and which ends up killing its owner. In the 1940s, the American science fiction writer Isaac Asimov wrote a series of stories about robots and invented the term robotics, the science of robots. Meanwhile, in the real world, the first robots were developed by an engineer, Joseph F. Engelberger, and an inventor, George C. Devol. Together they started a manufacturing company that produced the first real robot in 1961. Robots of this type were installed at automobile plants and proved to be a success. They worked reliably and saved money for companies.

(2) These industrial robots were nothing like the terrifying creatures that can often be seen in science fiction films. Since they were made of metal, they could perform certain jobs that were difficult or dangerous for humans, particularly jobs that involve high heat. And since robots were tireless and never got hungry, sleepy, or distracted, they were useful for tasks that would be tiring or boring for humans. Industrial robots have been improved over the years, and today they are used in many factories around the world.



(3) Though the use of robots has meant the loss of some jobs, at the same time other jobs have been created in the design, development, and production of the robots. Outside of industry, robots have also been developed and put to use by governments and scientists in situations where humans might be in danger. For example, they can be sent in to investigate an un-exploded bomb or an accident at a nuclear power plant. Researchers also use robots to collect samples of hot rocks or gases on active volcanoes. In space exploration, robots have performed many key tasks where humans could not be present, such as on the surface of Mars. In 2004, two robotic Rovers—small six-wheeled computerized cars—were sent to Mars. The Rovers had lasers that functioned as eyes and software that was designed to help them travel around holes or rocks. While on Mars, they performed many scientific experiments and took thousands of photographs, sending all the results back to Earth.

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(6) Robots have come a long way. Once, story writers and movie-makers imagined them as dangerous humans, like machines that could take over the Earth. Now, robots help people at home, in factories, and in scientific research. In the next decade, many new uses will be found for robots, especially in cars. As one designer said, "A person's role in the car is changing. People will become more planners than drivers. Robots will do the rest".

Complete the sentence with most suitable alternative.

In industry, robots.....

have taken away some jobs and created others.

A-

have caused many factories to close.

B-

look and behave a lot like human beings.

C-

are used instead of computer programmers and engineers.

D-

work only

E-

Q195

In the question below, the sentences have been given in Indirect speech.

From the given alternatives, choose the one which best expresses the given sentence in Direct speech.

His father ordered him to go to his room and study.

His father said, "Go to your room and study."

A-

His father said to him, "Go and study in your room."

B-

His father shouted, "Go right now to your study room"

C-

His father said firmly, "Go and study in your room."

D-

His father said politely, "Go and study in your room."

E-

Q196

Read the following passage and answer the question that follows.

The Indian middle class consists of so many strata that it defies categorization under a single term class, which would imply a considerable degree of homogeneity. Yet two paradoxical features characterize its conduct fairly uniformly; extensive practice and intensive abhorrence of corruption.

In the several recent surveys of popular perceptions of corruption, politicians, of course, invariably and understandably top the list, closely followed by bureaucrats, police officers, lawyers, businessmen and others. The quintessential middle class. If teachers do not figure high on this priority list, it is not for lack of trying, but for lack of opportunities. Over the years, the sense of shock over acts of corruption in the middle class has witnessed a steady decline, as its ambitions for a better material life have soared, but the resources for meeting such ambitions have not kept pace.

What is fascinating, however, is the intense yearning of this class for a clean, corruption-less politics and society, a yearning that has again and again surfaced with any figure, public or obscure, focusing on his mission of eradicating corruption. Even the repeated failure of this promise on virtually every man's part has not subjected it to the law of diminishing returns.

The intense Middle Class intensely yearns for

- A- better material resources
- B- extensive practice of corruption
- C- clean honest society
- D- law of increasing returns
- E- freedom of speech

Q197

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Teachers are not high on the list of corruption because they do not have

- A-Courage
- B-Opportunities
- C-Support
- D-Ambition
- E-Willingness

Q198



Read the following passage and answer the question that follows.

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In the several recent surveys of popular perceptions of corruption, politicians, of course, invariably and understandably top the list, closely followed by bureaucrats, police officers, lawyers, businessmen and others. The quintessential middle class. If teachers do not figure high on this priority list, it is not for lack of trying, but for lack of opportunities. Over the years, the sense of shock over acts of corruption in the middle class has witnessed a steady decline, as its ambitions for a better material life have soared, but the resources for meeting such ambitions have not kept pace.

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The Indian Middle class is

- A-Defiant
- B-Mysterious
- C-Homogeneous
- D-Stratified
- E-Unique

Q199

Read the following passage and answer the question that follows.

The greatest thing this age can be proud of is the birth of man in the consciousness of men. In his drunken orgies of power and national pride, man may flout and jeer at it. When organized national selfishness, racial antipathy and commercial self-seeking begin to display their ugly deformities in all their nakedness, then comes the time for man to know that his salvation is not in political organizations and extended trade relations, not in any mechanical re-arrangement of social system but in a deeper transformation of life, in the liberation of consciousness in love, in the realization of God in man.

In this passage, what does the phrase "God in man" imply?

- A- God having assumed the shape of man
- B- Neither fully godly nor fully human
- C- Man being transformed into God
- D- The divine qualities in man
- E- the divine qualities of God

Q200

Read the passage and answer the question based on it.

Worldwide illiteracy rates have consistently declined in the last few decades. One of the major reasons for this decline is the sharp increase of literacy rates among young women, which is a result of specific campaigns designed to increase educational opportunities for girls. However, there are still an estimated 771 million illiterate adults in the world, about two-thirds of who are women.

Based on the above passage, the author would tend to agree with which of the following statements?

- A- Men and women should have equal access to education.
- B- Men have a greater need for higher education than women.

- Worldwide, women need medical care more than the ability to read.
- C- It has been proven that women with increased education have fewer children.
- D- Women are having all opportunities.
- E-

Answer

1-A

Logic:

Here 5 is repeating after every 2nd number

28 25 5 21 18 5 14

Other numbers getting reduce by 3 and 4 as marked in the series

So next term is 11,5

Reducing 14 by 3 and 5 is getting repeated after every 2 term

2-B

Here there is 2 series as marked

13 29 15 26 17 23 19

So reduce 23 by 3 to get 20

a increase 19 by 2 to get 21

So next terms are 20,21

3-E

Eligibility Criteria for Medical Officers:

The applicant must:

1. Hold an MBBS degree with at least 50% marks.
2. Have a minimum of 4 years of full-time rural practice experience.
3. Be willing to sign a 3-year bond of service.
4. Have good knowledge of the local language.

Special Consideration Clause:

If an applicant satisfies all other criteria except one, the case is to be examined further (though the continuation of the sentence is cropped off in the image).

Given Criteria Recap:

An applicant must:

- 
1. **MBBS degree with minimum 50%**
 2. **Minimum 4 years of full-time practice in rural areas**
 3. **Willing to execute a 3-year service bond**
 4. **Good knowledge of local language**

Gauri's Profile:

- MBBS with **47%** (does **not meet** the 50% MBBS minimum)
- MD with **56%**
- 5½ years rural experience
- Willing to execute 3-year bond
- Good knowledge of local language and Hindi

Which condition does she fail?

- She fails at (i) → MBBS marks < 50%

BUT

Per **Option C:**

(C) At (i) above, but has a minimum of **45% marks at MBBS** and has done **MD/MS** with at least **50%**, be referred to the **Chief Medical Officer (C.M.O.)**

Gauri has:

- **47% in MBBS** ($\geq 45\%$)
- **56% in MD**

Conclusion: Gauri's case should be referred to the Chief Medical Officer

4-A

Selection Criteria Recap:

The applicant must:

1. Hold **MBBS degree with minimum 50%**
2. Have **minimum 4 years rural practice**
3. Be **ready to sign a 3-year bond**
4. Have **good knowledge of the local language**

Payal's Case Summary:

- Has MBBS and M.S.
- Practiced in native village for **5 years**
- Knows **local language well**
- Has **>60% in all exams**
- **Willing to sign 3-year bond**

Conclusion:

Payal **satisfies all the required conditions.**

Thus, she **should be selected directly** — no referral is necessary.

Final Answer: **SELECT** the candidate.

5-A

Full Evaluation of Raghav:

1. **MBBS degree with minimum 50%**
→ *Not explicitly mentioned*, but no mention of low marks implies likely $\geq 50\%$

2. **Minimum 4 years of full-time rural practice**
He worked 6 years in a big city, so this is **not fulfilled**
3. **Willing to sign 3-year bond**
4. **Good knowledge of local language**

So, he fails only at (ii) — rural experience.

Check if Exception (A) Applies:

Option A: At (ii) above, but has **4 years full-time in urban/semi-urban**, and spent **5 years in rural areas any time during life** → refer to **Secretary of Trust**.

He has:

- 6 years urban practice
- Studied in rural area (meets “spent 5 years in rural area anytime”)

So **he satisfies Exception A**, hence **eligible under the exception clause**.

BUT the final line in the instruction says:

"In case the applicant who satisfies all other criteria **except one**, the case may be considered further..."

Final Key Point:

He meets **all criteria either directly or through the accepted exception (A)**.

So **he should be SELECTED**.

6-D



Here:

A-athletes

W-women teacher

P-play

In statement there is no information about males

So Conclusions anything on males , we cant say 100%

So false

And there is no intersection in play and athletes as you can see in image so false

Both conclusion false

7-B



Here:

M-mangoes
g-golden
C-cheap



There cant be relation ship between mango and cheap as shown in the figure because there is no relationship between golden and cheap
And mango is present in golden

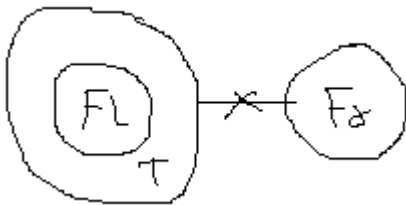
Conclusion 1 false

Conclusion 2 is true

Yes it is true no relation between golden colored mango and cheap

Reason explained above

8-E



Here:

Fl-flower

T-tree

Fr-fruit

Conclusion 1

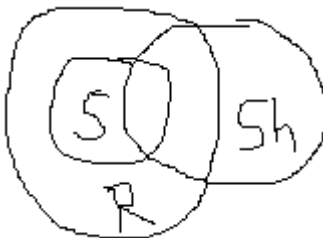
We can clearly see there is no relation between **flower and fruit**

So 1 is true

Conclusion 2

We can say trees are flower so it is also true

9-A



Here:

S-sword

R-rusty

Sh-sharp

Since statements are positive

Then all negative conclusions will be false

So conclusion 2 is false (its negative)
And we can see intersection between rust and sharp
So we can say some rusty things are sharp



10-C



Here there is 2 series as shown

First series:

$$8 + 3 = 11$$

$$11 + 3 = 14$$

$$14 + 3 = 17$$

And so on

Second series:

$$43 - 2 = 41$$

$$41 - 2 = 39$$

And so on ...

11-C

Neither Assumption A nor B is implicit.

Reason:

There is no direct connection between the *popularity of Ayurvedic medicines* and *India's glorious past*. The popularity could be due to various other factors such as effectiveness, government promotion, or changing health preferences — not necessarily because of the country's ancient history. Therefore, **both assumptions are not valid**.

12-B

Only Assumption B is implicit.

Explanation:

If someone says "*We must provide conveyance allowance to our employees,*" it implies that **employees should get a reward for being disciplined** (Assumption B). That is a valid assumption — it justifies the action.

However, **Assumption A** — "*Company should provide conveyance allowance because it is true*" — is **not implicit**. The statement does not give any reason or evidence that it is "true" or justified in general; it only shows what the speaker thinks *should* be done.

Thus, **only Assumption B is implicit**, and **Assumption A is not**.

13-A

Only Assumption A is implicit.

Explanation:

When the government decides to appoint 1,000 teachers, it assumes that there are enough primary schools where these teachers can be posted. So, **Assumption A is implicit**.

However, **Assumption B** — that *teachers are not available in the state* — is not necessarily assumed. The statement does not mention any shortage or unavailability of teachers; it only talks about a decision to appoint. So, **B is not implicit**.

14-D

Given:

- A is in the place of B after interchanging positions.
- There are **11 people to the left of B**.
- There are **8 people to the right of A** after the interchange.

To Find:

Total number of people in the row.

Explanation:

- After interchange, A is in B's original position.
- There are **11 people to the left of B** → So, after interchange, A also has 11 people to the left.
- There are **8 people to the right of A** → This is the new position (i.e., B's original place).

So, the total number of people in the row =

People to the left of A + A + People to the right of A

= 11 (left) + 1 (A) + 8 (right) – 1 (Bcz A is counted twice)

= **19**

15-C

CMM, EOO, GQQ, _____, KUU

Logic

C +2 we get E

E +2 we get G

G +2 we get I

I +2 we get K

Same logic for 2nd and 3rd elements of the series

ISS is the missing

16-B

4, 6, 12, 14, 28, 30, _____

Logic

4 +2 = 6

6 *2 = 12

12 +2 = 14

14*2 = 28

28 +2 = 30

Now,

30*2 = **60**

17-A

Step 1: Separate into two alternating series

- **Series 1 (odd positions):** 10000, 9900, 9801
- **Series 2 (even positions):** 11000, 10890, ?

Analyze Series 1:

10000 → 9900 → 9801

- 10000 → 9900: decrease by **100**
- 9900 → 9801: decrease by **99**

Pattern: Decreasing by 100, then 99 → likely next is **98**

9801 - 98 = 9703

Analyze Series 2:

11000 → 10890

- 11000 → 10890: decrease by **110**

If we follow this pattern:

Next decrease = 109

10890 - 109 = 10781

Final series:

10000, **11000**, 9900, **10890**, 9801, **10781**

Answer: 10781

18-B

Check **difference between the numbers**:

- A: 70–80 → **+10**
- B: 54–62 → **+8**
- C: 28–32 → **+4**
- D: 21–24 → **+3**
- E: 14–16 → **+2**

Now spot the pattern:

10, 4, 3, 2 → small differences

But B is **+8**, which is **not following a decreasing or clear sequence**

So, final conclusion:

Option B: 54–62 is the odd one out

- ✓ Difference of **8** doesn't fit with other differences
- ✓ No pattern in factors or parity that aligns with the others
- ✓ Digit sum and position don't follow the rest

19-D

37,14,19,7

Is the odd one out

14 is not the prime number

Logic

All numbers of other options are **prime number**

This not followed by option D bcz of 14 present

20-C

Logic

R-2 we get **P**

O+2 we get **Q**

A-2 we get **Y**

S+2 we get **U**

T-2 we get **R**

So its -2,+2,-2

Applying same for SLOPPY we get **QNMRNA**

21-A

Code logic is

health → GSKZDG

health → GSKZDG

h is coded as G and placed at last as shown in the figure
(h-1 is G),(e-1 is D),(a-1 is Z)..... and so on

Same for other alphabets

So code for north is **GSQNM**

22-D

BEAT coded in YVZG

Logic

Coded with **Opposite letters**

Letter opposite to B is Y , letter Opposite to E is V and so on

(English alphabet is divided into 2 parts, that is A to M and then N to Z)

So code for MILD is **NROW**

Explanation 23 to 25

	T – shirt	trouser
A	Black	Red
B	Red	White
C	Violet/White	Yellow
D	Green	Indigo
E	Yellow	Blue
F	White/Violet	Black
G	Blue	Cream

23-B

G wear cream colour trouser

24-C

Either C or F wears violet t shirt



25-B

B wear white trouser

26-C

1. **Poisonous spider** → **Agnoscrel**
2. **Brown spider** → **Agnos deery**

Step 1: Find common words across the statements

We want to **find the code for "spider"**

- "spider" is present in **all 3 phrases**
- So the code for "spider" must be the **common word** across all 3 codes

Let's check the code words:

- Sentence 1: **Agnos, vitri, blumin**
- Sentence 2: **Agnoscrel**
- Sentence 3: **Agnos, deery**

Now observe:

- **"Agnos"** appears in Sentence 1 and 3
- But Sentence 2 has **"Agnoscrel"**, which is similar — could be a **compound or extended form** of "Agnos"

So likely:

"Agnos" = spider

Step 2: Verify

- In sentence 3: **"Brown spider" → Agnos deery**
→ If **"Agnos"** = spider, then **"deery"** = brown
- In sentence 1: **Black Widow spider → Agnos vitri blumin**
→ So **"vitri"** and **"blumin"** could mean **Black** and **Widow**, order unknown

Explanation 27 to 30

Players	Home Town	Specialist
Sumeet	Pune	Captain
Philips	Warangal	Spinner
Wasim	Chandigarh	Batman
Bishan	Surat	Pace Bowler
Chetan	Bangalore	Wicket Keeper

27-A

Spinner is Philips

28-C

Spinner hometown is Warangal

29-B

Chandigarh is a home town of Wasim

30-D

Pace bowler is Bishan

31-D



migenlasan means cupboard
lasanpoen means boardwalk
cuopdansa means pullman

By using first 2 statements

Code for walk is **poen** (by taking out common term)

But we cant find code for way from any of the given statements

So by seeing options and after elimination we can say that code for way can be **forc** it's a possibility

Explanation for 32-34

Weights (Note = * is Multiplication)

- A **$2*4.5*0.5*X$**
- B **$4.5*0.5*X$**
- C **$0.5*X$**
- D **X**
- E **$2*X$**

Here we can assume any weight of E (example – 10kg etc)

Or assume any variable (a,b,c,...) in terms of weight

And then find rest of the weights as per conditions given in the question

Here we have assumed weight of **E** as **2X** and found rest of the weight in terms of **X**. As shown in the figure above

32-C

Lightest in weight is **C**

33-A

E is lighter than **A,B**

34-B

E is heavier than **D,C**

35-B

MAIN word will form

And I is 3rd letter

Explanation for 36-38

Administration: F , G

Accounts: C , A , B/E

Operation: H , B/E



Note

G,C,H are females rest are males

Earnings

$G > H > A > \underline{F/B/E} > C$

Note

In **F/B/E** we don't know ranking of there earning who earns more or less is clue less

That way / is used

But we know **F,B,E** less then **G,H,A** and more then **C**

36-B

3 people works in Accounts

37-B

Females are G,C,H

38-C

A rank 3rd in terms of earning

39-B

Row 3:

$108 \rightarrow 12 \rightarrow 18$

Check pattern:

- $12 \times 1.5 = 18$
- $108 \div 9 = 12$

Row 2:

- $96 \div 6 = 16$
 - $16 \times 0.75 = 12$
- This matches

Row 1:

- $72 \div 3 = 24$
- $24 \times 0.25 = 6$

So:

Pattern in each row:

- First $\div$ some number = middle
- Middle $\times$ ratio = last

This ratio changes:

- Row 1: $\times 0.25$
- Row 2: $\times 0.75$
- Row 3: $\times 1.5$

Pattern: $0.25 \rightarrow 0.75 \rightarrow 1.5$ (increasing $\times 3$)

So it fits!

Final Confirmation:

- $108 \div 9 = 12$

- $12 \times 1.5 = 18$

✓ Matches perfectly.



40-A

- The number of people who belong **only to Chess** is **6**.
- The number of people who belong **only to Drama** is **3**.
- The number of people who belong **only to Art** is **2**.
- Some people belong to two groups such as:
 - Chess and Drama (but not Art): **2** people
 - Drama and Art (but not Chess): **3** people
 - Chess and Art (but not Drama): **2** people
- Additionally, **5** people belong to **all three groups**.

Now, we are asked: **How many people belong to exactly one group?**

To answer this, we consider only those who are part of **only one of the groups**, not any overlaps.

These are:

- People who are only in Chess: **6**
- People who are only in Drama: **3**
- People who are only in Art: **2**

Adding them gives:

$$6 + 3 + 2 = 11$$

Explanation for 41-44

Tuesday: U (Oncologist)

Wednesday: Q (ENT) , R (Paediatrician)

Friday: S (Radiologist) , T (Neurologist)

Saturday: V (Orthopedics) , P (Ophthalmology)

41-D

On Saturday – orthopedics and ophthalmology visits

42-B

S is Radiologist

43-B

P is Ophthalmology

44-C

T visit on Friday

45-A

Here there are 2 series as shown



Logic

$$21 + 8 = 29$$

$$29 + 8 = \mathbf{37}$$

$$25 + 8 = 33$$

$$33 + 8 = \mathbf{41}$$

And **18** is repeated after every 2 numbers

46-D

Using 1 we can say $C/E > D$

Using 2 we can say $B > A$

Combining 1 and 2 also we cant say, who secured last

47-C

Statement 1 say Q is right of P and left of S

So we can say

R is opposite to S and is partner as well

Statement 2 says - S is opposite to P

It means we can say Q and R are opposite and are partners also

So either of the statement is enough to get answer

48-E

Statement 1: 2,3,5,7.....(prime numbers)

Statement 2: 2,4,6,..... (even numbers)

Combining both the statement we get 1 common date which is 2 Feb

Hence got the birth date

So both required

49-B



1. **343 cubelets** means the cube is divided into

$$\sqrt[3]{343} = 7$$

So, the original cube is divided into a $7 \times 7 \times 7$ grid of small cubes.

2. To divide a cube into **7 parts** along each dimension, you need **6 cuts** (because:

- 1 cut gives 2 pieces,
- 2 cuts give 3 pieces,
- So to get 7 parts, we need 6 cuts per direction).

3. Since the cube has **3 dimensions** (length, breadth, and height):

- We need **6 cuts along each of the 3 dimensions**.

So, the **minimum number of cuts made is 18**.

50-C

We are given:

- A cube is divided into **343 identical cubelets**, which means $343 = 7^3$, so the cube is divided into a $7 \times 7 \times 7$ grid.
- Before dividing, the **outer cube is painted**:
 - One **set of adjacent faces** is painted **green**,
 - Another set is painted **red**,
 - The third set is painted **blue**.
 - So, the cube has **three pairs of opposite faces**, and **each adjacent pair** (like front-top, front-left, etc.) is painted with a different color pair.

We are to find **how many cubelets are painted with exactly two colors**.

- Cubelets with **exactly two colors** lie along the **edges**, but **not at the corners**.
- There are **12 edges** in a cube.
- Each **edge** of the cube has **7 cubelets**.
- On each edge:
 - The **two ends (corner cubelets)** are painted with **three colors**.
 - The **middle 5 cubelets** (since $7 - 2 = 5$) are painted with **exactly two colors**.

So,

$$\text{Number of cubelets with exactly two colors} = 12 \text{ edges} \times 5 \text{ cubelets per edge} = \boxed{60}$$

51-D

There is no relationships between them

Statement 1 and Statement 2 are effects of independent causes

There is 2 different happening and there are 2 different effects of those happening

52-B

Statement 2 – parent protested against high fee (this is happened or cause)

Now

Due to this protest, government fixed the fees for professional courses .

This is nothing but statement 1

So we can say statement 1 (fixed fees) was the effect of (protest) which is nothing but statement 2

53-C

Both statements are individual happenings or causes

They are not related

54-B

Asian countries are importing steels

So its means companies manufacturing steel will make profit

So statement 2 happened or caused (importing steel)

And the effect of its was seen in terms of **profit made**

So statement 1 is effect

55-C

- Player **seeded 43** won the tournament.
- We are to find who **cannot** be the **runner-up**.
- Only players in the **opposite half of the bracket** can be runner-up.
- Seed **43** played in Match 22 (Seed 22 vs Seed 43) ⇒ belongs to the **bottom half** of the bracket (Matches 17 to 32).
- Thus, seeds involved in Matches 17 to 32 are in the **same half** as Seed 43.

These include seeds from approximately **17 to 48**, because:

- Match 17: Seed 17 vs 48
- Match 18: Seed 18 vs 47
- ...
- Match 32: Seed 32 vs 33

So, **Seed 46** would also be in this half of the draw.

Conclusion:

If **Seed 46** is in the **same half** as **Seed 43**, they would have met **before the final**, so **46 cannot be the runner-up**.

Correct Answer: Seed 46 cannot be the runner-up.

56-B

Step 1: Match Format (Round 1 Pairings)

In a knockout tournament of 64 players:

- Match 1: Seed 1 vs 64

- Match 2: Seed 2 vs 63
- Match 3: Seed 3 vs 62
- ...
- Match 12: Seed 12 vs 53
- ...
- Match 32: Seed 32 vs 33

We observe that:

- Player seeded **12** plays against **53** in Match 12
- So Seed **12** is in the **top half** of the bracket (Matches 1–16)

That means the opponent in the final must be someone from the **bottom half** (Matches 17–32).

Step 2: Who is in the Bottom Half?

- Match 17: Seed 17 vs 48
- Match 18: Seed 18 vs 47
- ...
- Match 32: Seed 32 vs 33

So seeds like **17 to 32** and **33 to 48** are in the bottom half.

But notice the **pairings**: High vs Low seeds always match to total 65.

That is:

Match i : Seed i vs Seed $(65 - i)$

So:

- Match 2: Seed 2 vs 63
- Match 3: Seed 3 vs 62
- ...
- Match 16: Seed 16 vs 49

So Seed **63** plays in Match 2 → meaning it's in the **top half**, right?

Yes, but now here's the twist:

The way the knockout bracket works, the **seed number alone doesn't determine the half** — it's about who survives from the **opposite branch of the tree**.



Step 3: How can Seed 63 reach the final?

Let's look at the structure:

- Seed 12 is in top half (Match 12)
- For a player to reach the final and face them, they must come from the **bottom half of the tree** (i.e., opposite half in terms of final bracket path)

Now the **key** idea is: regardless of whether Seed 63 *started* in top half, we can construct a **path** where:

- Seed 63 keeps winning
- Eventually reaches the final from the **opposite side**

Because **seeding doesn't strictly dictate the top/bottom half path** — it's about the structure of **winners progressing** and the way the bracket connects.

So it's **theoretically possible** for **Seed 63 to face Seed 12 in the final**, provided they are on opposite sides of the knockout tree.

Final Answer:

Seed 63 can be the lowest seeded player to reach the final and face Seed 12.

Correct answer: 63

57-A

Step 1: Understand the structure of a knockout tournament

This is a **single-elimination knockout** of 64 players, with 6 rounds:

- Round 1: 64 → 32 players
- Round 2: 32 → 16 players
- Round 3: 16 → 8 players
- Quarterfinals → 4
- Semifinals → 2
- Finals → 1 winner

So, for **two players to face each other**, their paths in the tournament must intersect — i.e., they must be in the same branch and meet in a round.

◆ Step 2: Determine in which round players 23 and 46 could meet

In Round 1:

- Match 23 = Seed 23 vs Seed 42

So, **seed 23 and 46 do not meet in Round 1**

But, if **seed 23 wins Match 23** and **seed 46 wins Match 18 (vs 47)**, their paths might meet later.

Let's simulate a potential path:

- **23 wins Match 23** (vs 42)
- **46 wins Match 18** (vs 47)

Now:

- In Round 2:
 - Winner of Match 23 (Seed 23) vs Winner of Match 10 (Match 23 is matched with Match 10)

Wait: we need to know how **match numbers pair** into second round.

Per the passage:

- 2nd Round Match 1 = Winner of Match 1 vs Winner of Match 32
- 2nd Round Match 2 = Winner of Match 2 vs Winner of Match 31
- ...

So, Match i pairs with Match $(33 - i)$

Hence:

- Match 23 pairs with Match 10 (since $33 - 23 = 10$)

That means **Seed 23 (Match 23)** must face **Winner of Match 10** in Round 2.

Match 10 = Seed 10 vs 55

→ So **Seed 23 and 46 will NOT meet in Round 2**

Now check Round 3:

- Match 23 → Winner goes to Round 2 → then Round 3

We eventually find that **Seed 23 and 46 could meet in Round 3 or 4**.

So yes — it's **possible** for seeds like **23 and 46 to meet in later rounds**.

◆ Step 3: What does this imply for the next part?

The question asks:

If 23 and 46 can meet, then **which other pair could also be possible?**

We need to **choose a pair** who could also be on a **possible intersecting path**.

Given Options (assumed):

- (9, 13) ← Correct
- (5, 60)
- (4, 62)
- ...

Let's analyze **Seeds 9 and 13**

Round 1:

- Match 9: Seed 9 vs 56
- Match 13: Seed 13 vs 52

→ So they're both **not in the same match**, but are in **same half of the bracket**.

Check pairing in Round 2:

- Match 9 pairs with Match 24 ($33 - 9 = 24$)
- Match 13 pairs with Match 20 ($33 - 13 = 20$)

So Seed 9 could face Seed 13 in **Round 3 or 4**

Hence, **Seeds 9 and 13 can meet in a later round** — just like Seeds 23 and 46.

Final Answer:

Players seeded 9 and 13

These two players can logically meet in the tournament, **just like 23 and 46**, so the answer is valid.

58-C



Logic

$$B + 3 = E$$

$$F + 3 = I$$

Same logic for other alphabets

$$R + 3 = U$$

$$B + 3 = E$$

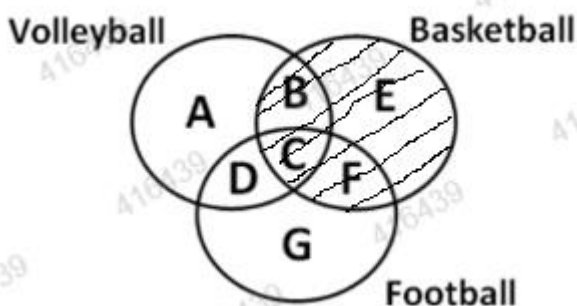
$$G + 3 = J$$

$$J + 3 = M$$

59-B

We want intersection of Volleyball or Football but not basketball

A+D+G is that region



We don't want shaded part and yes rest part which is unshaded is our answer

60-E

We are given:

- Total number of girls = 26
- After moving 4 places to the left, Sakshi is 10th from the left end

We are asked to find:

→ Her earlier position from the right end

1. Sakshi's new position from the left = 10
2. She moved 4 places left, so her earlier position from the left =
 $10 + 4 = 14$
3. Now, her position from the right =
 $\text{Total} + 1 - \text{Position from left} = 26 + 1 - 14 = 13$

13th from the right end

61-A

After de coding

R and Q is couple , Q is male

P is the son of couple

So we can say Q is the father of P

62-E

After de coding

Q is the parent of R,Q

And R,P are brothers

We can say R,P are sons of Q

63-A

After de coding

P is the sister of R and R is brother of P and they are children of Q

64-B

The boy playing is the brother of Mohan

65-C

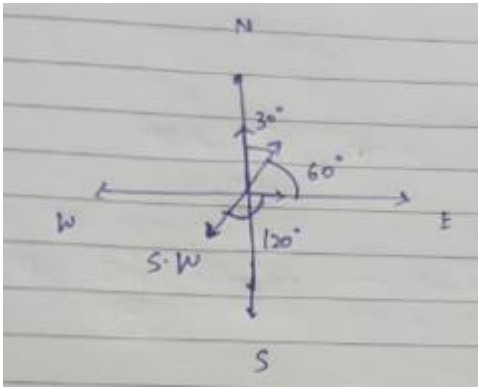
Kalyani (female) and Dheeraj are couples

There children's are Ashok and Sudeep(male)

Sudeep is married to veena

And we don't know gender of Ashok

66-A



So man looking at south-west then he moves 210 degrees in anti clock wise direction so he is facing north

210 is covered in this manner (120+60+30) as shown in the figure

67-D

Step 1:

He walks **12 km north** from point A.

Step 2:

He **turns right** → now facing **east**, and walks **4 km**.

Step 3:

He **again turns right** → now facing **south**, and walks **9 km**.

Now, calculate his final position relative to the starting point:

- **North–South direction:**
 - He first goes 12 km **north**, then 9 km **south** → net = **3 km north**
- **East–West direction:**
 - He goes 4 km **east**

Final Position:

From the starting point, he is:

- **3 km north**
- **4 km east**

Use the **Pythagoras theorem** to find the straight-line distance:

$$\text{Distance} = \sqrt{(3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ km}$$

Direction:

He is **North-East** of the starting point.

68-B

Given:

- **A fires 5 shots for every 3 shots of B**
- **A hits (kills) once in every 3 shots**
- **B hits (kills) once in every 2 shots**
- **B has missed 27 times**

Step 1: Find total shots fired by B

B's accuracy:

B kills once in every 2 shots → So, B misses once in every 2 shots too.

That means, in 2 shots, B gets 1 hit and 1 miss.

So, if B has missed 27 times, then:

$$\text{Number of 2-shot groups} = 27 \text{ misses} = 27 \text{ groups}$$

$$\text{Total shots by B} = 27 \times 2 = 54 \text{ shots}$$

Step 2: Find total shots by A

A fires 5 shots for every 3 shots by B

$$A's \text{ shots} = \left(\frac{5}{3}\right) \times 54 = 90 \text{ shots}$$

Step 3: A's kill rate

A kills once in every 3 shots.

$$A's \text{ kills} = \frac{90}{3} = 30 \text{ kills}$$

69-B

We're given:

- A printer uses **3189 digits** to number the pages of a book starting from page **1**.
- We are to **find the number of pages** in the book.

Let's break it down by number of digits per page:



1-digit page numbers:

Pages 1 to 9 $\rightarrow$ 9 pages

Digits used = $9 \times 1 = 9$

2-digit page numbers:

Pages 10 to 99 $\rightarrow 99 - 10 + 1 = 90$ pages

Digits used = $90 \times 2 = 180$

Total so far:

$9 + 180 = 189$ digits

3-digit page numbers:

Pages 100 to 999 $\rightarrow 999 - 100 + 1 = 900$ pages

Digits used = $900 \times 3 = 2700$

Total so far:

$189 + 2700 = 2889$ digits

We still have:

$3189 - 2889 = 300$ digits left

4-digit page numbers:

Each page uses 4 digits.

So, $300 \div 4 = 75$ pages

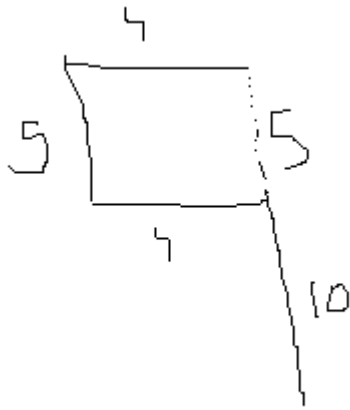
Final Count:

- 9 pages (1-digit)
- 90 pages (2-digit)
- 900 pages (3-digit)
- 75 pages (4-digit)

Total pages = $9 + 90 + 900 + 75 = 1074$

70-D

He is 15 km away from starting point in north direction as shown in the figure



71-C

To determine the last step for the given input "**in this volume provides the**", we first need to identify the pattern used in the example rearrangement steps.

Given

Input:

before its court seeking permission to carry out a test.

Observing the steps:

Step I:

its before court seeking permission to carry out a test.

→ First word alphabetically ("before", "its", "court", etc.) is "its" → moved to front.

Step II:

its before carryout court seeking permission to a test.

→ Next smallest word alphabetically among the remaining is "carryout" → placed after "its before".

Step III:

its before carryout court permission seeking to a test.

→ "permission" is placed next.

Step IV:

a its before carryout court permission seeking to test.

→ "a" inserted in the correct position.

Step V:

a its before carryout court permission seeking test to.

→ Final arrangement.

Key Observations:

- Words are being rearranged in **alphabetical order** from left to right.
- Each step moves one word to its appropriate position in the growing sorted list.

New Input:

Input:

"in this volume provides the"

Words in the input:

["in", "this", "volume", "provides", "the"]

Alphabetically sorted:

["in", "provides", "the", "this", "volume"]

Step-by-step arrangement:

- **Step I:** in this volume provides the

- **Step II:** in provides this volume the
- **Step III:** in provides the this volume
- **Step IV:** in provides the this volume (*no change as "this" and "volume" are already in place*)

But we are still not in full alphabetical order, as "this" should come after "the".

Correct order should be:

in provides the this volume

That means all 5 words are now sorted alphabetically.

Step III is the last step for the input **"in this volume provides the"**.

72-C

Input:

these many other clinicians and

Words:

['these', 'many', 'other', 'clinicians', 'and']

Sorted order:

['and', 'clinicians', 'many', 'other', 'these']

Step 0 (Input):

these many other clinicians and

Step I:

Move "and" to the front

→ **and these many other clinicians**

Step II:

Next word alphabetically: "clinicians"

→ **and clinicians these many other**

Step III:

Next word alphabetically: "many"

→ **and clinicians many these other**

That matches the option mentioned in the question.

So step 3rd

73-B

Logic explained in above question

Schooling play vital role in

74-A

a becoming modern man is chronic victim

75-C

Here we have got 3 pairs

having **same number of digits as in Ascending also**

Pair : (5,3),(3,8) and (7,4)

76-A

Hint – here groups are been formed by specific feature. which a figure is having

1,7,9 form a group bcz. a smaller fig of it self is embedded in that figure itself

2,3,6 forms a group bcz. same smaller fig is attached on side

4,5,8 forms a group bcz fig is divided into equal parts



77-A

Hint – see the rotation of arrow.(180 Degree)

And dots are getting increase from 1 fig to another

So opt A. is the answer

78-D

U is rotating in clock wise direction in 45 degrees

And triangle is placed alternatively in question figure

And dot remains the same

So opt D

79-B

See the rotation of rectangle

And black dot and white dot is placed alternatively

In 2 black dot figure there is 3 white dot

In 3 black dot figure there is 2 white dot

And this logic is alternatively followed

80-A

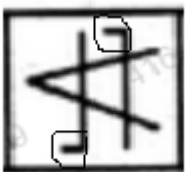
Fig 1 is odd one out

Bcz lollipop is present of vertex

And in rest it is present on side .

81-D

Logic - The L shaped fig is looking at same direction as shown in figure



Both the marked part is facing **same direction** that is **west**

And for figures marked part is facing **each other** that is **east and west**

82-E

Fig 5 is odd one out



The row is divided in to multiple parts as shown in the figure

Logic is - **sum of parts of top row and middle row is equal to parts of bottom row.**

This logic is followed in **first four figures**

Fig 5 is not following the same logic

Bottom row should consist of 8 parts but 9 parts are present



83-E

Hint - the star having dot in the middle is repeating after every two figures

And dot on vertex of star is increasing

So option E

84-C

Blue and black part is moving up and down from one figure to another figure

So opt C

85-C

86-C

See the rotation of black dot it moves in clock wise direction

So opt C

87-A

See the movement of flame of a candle it follows a particular pattern

All are looking at same direction (in the single set of 3 candles)

And flame size is also increasing

So option A

88-A

89-C

Make a pair of 2 figures and then see the rotation of circles present into oval fig and also circles are increasing they are also in alternatively in black and white circles

So opt C

90-B

91-B

Logic- A circle having no elements in it will come next

92-A

See the rotation of square

It moves from 1 corner to another in anti clock wise direction

Middle dot comes after every 2 figure

93-D

Rotation of **U** is in anti clock wise direction

And triangle is placed alternately

And dot moves in anti clock wise direction



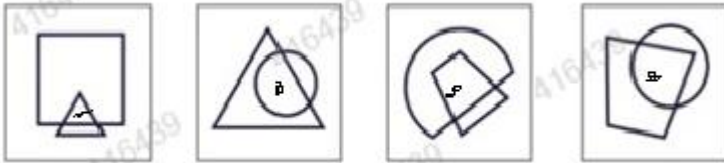
94-E

Black circles are placed 5,7,5....

In this pattern alternately in the series

So after 7 black circles , figure which will come will be having 5 black circles

95-B



Logic:

The intersection part as shown in the figure by a **dot**, that figure (intersection part) becomes the part of next figure

Example:

In figure 1 **triangle** is formed by intersection and triangle is seen in the next figure and same for rest of the series

96-A

Here series is getting repeated after every 4 fig

Bcz fig 5 is same as fig 1

Therefore opt **A** is the answer

Bcz opt **A** comes after fig 1

Logic:

Basically fig 5 becomes fig 1

Fig 6 will become fig 2

Fig 7 will become fig 3 and so on

97-D

Hint – see the rotation of elements and eliminate options

Example - + moves in anti-clock wise direction from 1 fig to another

Heart also moves in anti clock wise direction

98-C

Option **C** is the odd one out

Bcz other options are **perfectly symmetric vertically** or they are **mirror image** of each other

99-B

Here grouping is done based on specific features of a figure

Here 2,3,9 becomes one group (geometrical figure is cut by a line)

Only one option is having 2,3,9 that is option2

Others options can be eliminated

100-B

Here 1,7,8 becomes one group (normal geometrical figure)
 3,5,6 becomes 2nd group (a straight line drawn in fig.)
 2,4,9 becomes 3rd group (fig divided in 4 parts)



101-C

We are given that a number is divided successively by 3, 4, and 7, yielding remainders 2, 1, and 4 respectively. We are to find the remainder when the same number is divided by 84.

Step-by-step breakdown:

Let the number be NNN.

When a number is divided **successively** by 3, 4, and 7 with remainders 2, 1, and 4:

- Let's work **backwards** to reconstruct the number.

Step 1: Last division is by 7, remainder = 4

So, let:

$$x \equiv 4 \pmod{7} \Rightarrow x = 7k + 4$$

Step 2: Before that, it was divided by 4, remainder = 1

So, we have:

$$x = 7k + 4 \Rightarrow \text{This was the result after division by 4, with remainder 1}$$

$$\begin{aligned} \text{So, } 7k + 4 &\equiv 1 \pmod{4} \Rightarrow 7k \equiv -3 \equiv 1 \pmod{4} \Rightarrow 3k \equiv 1 \pmod{4} \Rightarrow k \equiv 3 \pmod{4} \text{ (since } 3 \times 3 = 9 \equiv 1 \pmod{4}) \\ \Rightarrow k &= 4m + 3 \Rightarrow x = 7k + 4 = 7(4m + 3) + 4 = 28m + 21 + 4 = 28m + 25 \end{aligned}$$

Step 3: Before that, it was divided by 3, remainder = 2

So, again,

$$x = 28m + 25 \Rightarrow \text{This was the result after division by 3, with remainder 2}$$

$$\Rightarrow 28m + 25 \equiv 2 \pmod{3} \Rightarrow 28m \equiv -23 \equiv 1 \pmod{3} \text{ (since } -23 \equiv 1 \pmod{3}) \Rightarrow m \equiv 1 \pmod{3} \Rightarrow m = 3n$$

Final Expression:

$$N = 84n + 53$$

So, when this number is divided by 84, the **remainder is 53**.

102-A

Solution



- Total seats in the stadium = 1 (assume 1 for easy calculation).
 - $\frac{1}{4}$ of all seats are on the **lower deck**.
 - $\frac{4}{5}$ of the **lower deck** seats were sold.
 - $\frac{2}{3}$ of **all** the stadium seats were sold.
-

Step 1: Calculate unsold seats on lower deck

$$\text{Lower deck seats} = \frac{1}{4}$$

$$\text{Sold} = \frac{4}{5} \times \frac{1}{4} = \frac{1}{5}$$

$$\text{So, unsold lower deck seats} = \frac{1}{4} - \frac{1}{5} = \frac{5-4}{20} = \frac{1}{20}$$

Step 2: Total unsold seats in stadium

$$\text{Total sold} = \frac{2}{3}$$

$$\text{So, total unsold} = 1 - \frac{2}{3} = \frac{1}{3}$$

Step 3: Required fraction:

$$\text{Fraction of unsold lower deck seats to total unsold seats} = \frac{\frac{1}{20}}{\frac{1}{3}} = \frac{1}{20} \times \frac{3}{1} = \boxed{\frac{3}{20}}$$

103-B

1748 must be subtracted from 306452 to make it a perfect square. After subtracting we will get **304704** which is nothing but square of 552.

104-D

By using options we can say number is **364** which satisfy all the condition given in question

105-B

We are given three quantities of sugar:

- 184 kg
- 230 kg
- 276 kg

We need to divide these into the **minimum number of bags of equal size** without mixing.

This means we must find the **HCF (Highest Common Factor)** of the three quantities.



Step 1: Find HCF of 184, 230, and 276

Prime factorizations:

- $184 = 2^3 \times 23$
- $230 = 2 \times 5 \times 23$
- $276 = 2^2 \times 3 \times 23$

Common factor: $2 \times 23 = 46$

So, HCF = 46 kg

Step 2: Number of bags required:

- $\frac{184}{46} = 4$
- $\frac{230}{46} = 5$
- $\frac{276}{46} = 6$

Total bags = $4 + 5 + 6 = \boxed{15}$

106-B

Given:

- Average of **n students** = 64
- 8 new students join with average = 73
- New average becomes a **whole number**
- n lies between 25 and 60
- We need to **find n**

Step-by-step:

1. Total marks of old students = $64n$
2. Total marks of new students = $8 \times 73 = 584$
3. New total students = $n + 8$
4. New average = $\frac{64n+584}{n+8}$

This must be a **whole number**.

So:

$$\frac{64n + 584}{n + 8} \in \mathbb{Z}$$



Try $n = 36$:

- Total old marks = $64 \times 36 = 2304$
- New total marks = $2304 + 584 = 2888$
- New total students = $36 + 8 = 44$

$$\text{New average} = \frac{2888}{44} = 65$$

Here we have to do hit and trial or use options one by one

Here opt B – 36 satisfies

107-B

Given:

- Total players = 11
- Captain's age = 25
- Wicket keeper's age = $25 + 3 = 28$
- Excluding these two, the average age of the remaining 9 players = average age of whole team – 1

Let the **average age of the whole team** be x .

Step 1: Total age of whole team

$$\text{Total age} = 11x$$

Step 2: Total age of captain and wicket keeper

$$\text{Captain} + \text{WK} = 25 + 28 = 53$$

Step 3: Total age of remaining 9 players

$$\text{Remaining total} = 11x - 53$$

Step 4: Average age of remaining 9 players

$$\text{Average of 9 players} = \frac{11x - 53}{9}$$



Step 5: Given that this average is 1 less than the average of the whole team:

$$\frac{11x - 53}{9} = x - 1$$

Multiply both sides by 9:

$$11x - 53 = 9x - 9$$

X is 22

108-D

Given:

- Total pages = 512
- Average errors per page = 4
- Total errors = $512 \times 4 = 2048$
- Errors in first 302 pages = 1208
- Remaining pages = $512 - 302 = 210$

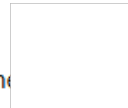
Step 1: Errors on remaining 210 pages:

$$\text{Remaining errors} = 2048 - 1208 = 840$$

Step 2: Average errors on remaining 210 pages:

$$\text{Average} = \frac{840}{210} = 4$$

109-B



We can solve this **backwards** from the final number of apples the woman has left after the customer.

Let's denote the apples remaining after each customer:

Step 1: Let final apples after 4th customer = 15

She gave away:

To 4th customer: $\frac{1}{2}$ of her apples + $\frac{1}{2}$

Let apples before 4th customer = x

Then:

$$x - \left(\frac{x}{2} + \frac{1}{2} \right) = 15 \Rightarrow \frac{x}{2} - \frac{1}{2} = 15 \Rightarrow \frac{x}{2} = 15 + \frac{1}{2} = \frac{31}{2} \Rightarrow x = 31$$

Step 2: Let apples before 3rd customer = y

$$y - \left(\frac{y}{2} + \frac{1}{2} \right) = 31 \Rightarrow \frac{y}{2} - \frac{1}{2} = 31 \Rightarrow \frac{y}{2} = 31.5 \Rightarrow y = 63$$

Step 3: Let apples before 2nd customer = z

$$z - \left(\frac{z}{2} + \frac{1}{2} \right) = 63 \Rightarrow \frac{z}{2} - \frac{1}{2} = 63 \Rightarrow \frac{z}{2} = 63.5 \Rightarrow z = 127$$

Step 4: Let apples before 1st customer = A

$$A - \left(\frac{A}{2} + \frac{1}{2} \right) = 127 \Rightarrow \frac{A}{2} - \frac{1}{2} = 127 \Rightarrow \frac{A}{2} = 127.5 \Rightarrow A = 255$$



Let the cost of each exercise book be x paise.

Then the total money the boy has is:

$$20 \times x = 20x \text{ paise}$$

If the price is reduced by 25 paise, the new price becomes:

$$x - 25 \text{ paise}$$

With this reduced price, he can buy 2 more books (i.e., 22 books) and still have 70 paise left. So:

$$22(x - 25) + 70 = 20x$$

Now solve:

$$22x - 550 + 70 = 20x \Rightarrow 22x - 480 = 20x \Rightarrow 2x = 480 \Rightarrow x = 240 \text{ paise}$$

Now calculate the total money:

$$20 \times 240 = \boxed{4800 \text{ paise}} = \boxed{₹48}$$

111-D

Let the two-digit number be represented as:

$$10x + y$$

Where:

- x = tens digit
- y = units digit

Now:

- Unit digit is increased by 100%, so it becomes $y + y = 2y$
- Tens digit is increased by 50%, so it becomes $x + \frac{1}{2}x = \frac{3}{2}x$



The new number becomes:

$$10 \times \frac{3}{2}x + 2y = 15x + 2y$$

The new number is 19 more than the original, so:

$$15x + 2y = 10x + y + 19$$

Now solve the equation:

$$15x + 2y = 10x + y + 19 \Rightarrow 5x + y = 19$$

Now try values of x (1 to 9) and see which y is valid (i.e., digit 0–9):

- If $x = 2$, then $5x = 10 \Rightarrow y = 9$

So, original number = $10x + y = 10 \times 2 + 9 = \boxed{29}$

112-C

☐ Original consumption = 10 kg

☐ Let original price be P

☐ So, original expenditure = $P \times 10 = 10P$

☐ Price increases by 32% $\rightarrow$ New price = $1.32P$

☐ The total expenditure increases by 10% $\rightarrow$ New expenditure = $1.10 \times 10P = 11P$

Now, let the new consumption be x kg. Then:

New price $\times$ New quantity = New expenditure

$$1.32P \times x = 11P$$

Cancel P from both sides:

$$1.32x = 11$$

$$x = \frac{11}{1.32} \approx 8.33 \text{ kg}$$

Result:

- New consumption = 8.33 kg
- Reduction in consumption = $10 - 8.33 = 1.67$ kg

113-B



Let the total number of residents be x .

According to the problem:

- 30% are men over 18 $\rightarrow 0.30x$
- 40% are women over 18 $\rightarrow 0.40x$
- The rest are children $\rightarrow x - (0.30x + 0.40x) = 0.30x$

We are told the number of children is 24, so:

$$0.30x = 24$$

$$x = \frac{24}{0.30} = 80$$

114-A

Given:

- **Original price** = ₹10
- **Original quantity sold** = 10 units
- So, **Original sales** = $10 \times 10 = ₹100$

Now:

- **Price increased to ₹11** (10% increase)
- **Quantity reduced to 8 units** (20% decrease)

New sales:

$$\text{New Sales} = 11 \times 8 = ₹88$$

Comparison:

Old sales = ₹100

New sales = ₹88

Change in sales = ₹100 - ₹88 = ₹12

$$\text{Percentage decrease in sales} = \frac{12}{100} \times 100 = 12\%$$

Conclusion:

- **Price increased by 10%**
- **Quantity decreased by 20%**
- **Sales revenue decreased by 12%**

115-B

Let the cost price of the first instrument be Rs. x , and the second instrument be Rs. $(8000 - x)$.

Given:

- First sold at 40% **profit** → Sale Price = $x + 0.40x = 1.4x$
- Second sold at 40% **loss** → Sale Price = $(8000 - x) - 0.40(8000 - x) = 0.6(8000 - x)$
- Both sale prices are **equal**:

$$1.4x = 0.6(8000 - x)$$

Solve the equation:

$$1.4x = 4800 - 0.6x$$

$$1.4x + 0.6x = 4800$$

$$2x = 4800 \Rightarrow x = 2400$$

116-A

Given:

- Total butter bought = **5 kg**
- Cost price = **₹300**
- So, cost price per kg = ₹60
- **1 kg is unsaleable**, so only **4 kg** is sold
- The **overall loss is 10%**

Step 1: Calculate Selling Price (SP) for overall loss

Loss of 10% on ₹300:

$$SP_{\text{total}} = 90\% \text{ of } 300 = \frac{90}{100} \times 300 = ₹270$$

Step 2: Selling price per kg

Since 4 kg is sold for ₹270:

$$SP \text{ per kg} = \frac{270}{4} = ₹67.5$$

117-A

We are given:

- Two chairs sold at Rs. 500 each.
- On one chair: **20% gain**.
- On the other chair: **12% loss**.

Let's calculate the **cost price** (CP) of each chair.

Chair 1 (20% Gain):

Selling Price (SP) = Rs. 500

Let CP be x

$$\text{Gain} = 20\% \Rightarrow 500 = x \times \left(1 + \frac{20}{100}\right) = x \times 1.2 \Rightarrow x = \frac{500}{1.2}$$

Chair 2 (12% Loss):

SP = Rs. 500

Let CP be y

$$\text{Loss} = 12\% \Rightarrow 500 = y \times \left(1 - \frac{12}{100}\right) = y \times 0.88 \Rightarrow y = \frac{500}{0.88}$$



Total CP and SP:

- Total CP = $416.67 + 568.18 = 984.85$
- Total SP = $500 + 500 = 1000$

Net Gain:

$$\text{Gain} = 1000 - 984.85 = 15.15$$

$$\text{Percentage Gain} = \frac{15.15}{984.85} \times 100 \approx \boxed{1.54\%}$$

118-C

We are given:

- The customer paid **Rs. 30.09**.
- Retailer made a profit of **25%**.
- Wholesaler made **20%** profit.
- Manufacturer made **18%** profit.

Let the cost price of the glasses to the manufacturer be xxx.

Step 1: Customer buys at Rs. 30.09 → Retailer earns 25%

Let cost price to retailer = R

$$R + 25\% \text{ of } R = 30.09 \Rightarrow 1.25R = 30.09 \Rightarrow R = \frac{30.09}{1.25} = 24.072$$

Step 2: Retailer bought at Rs. 24.072 → Wholesaler earns 20%

Let cost price to wholesaler = W

$$1.20W = 24.072 \Rightarrow W = \frac{24.072}{1.20} = 20.06$$

Step 3: Wholesaler bought at Rs. 20.06 → Manufacturer earns 18%

$$1.18x = 20.06 \Rightarrow x = \frac{20.06}{1.18} = \boxed{17}$$

119-B

We are given:

- A rectangular paper is folded into **two congruent parts**.
- When folded **along one set of sides**, the perimeter becomes **34 cm**.
- When folded **along the other set**, the perimeter becomes **38 cm**.

Let the original length be l and breadth be b .

Case 1: Fold along length $\rightarrow$ halves the length

Each folded part:

- Length = $\frac{l}{2}$, Breadth = b

Perimeter of folded part:

$$2 \left(\frac{l}{2} + b \right) = 34 \Rightarrow l + 2b = 34 \quad (\text{Eq 1})$$

Case 2: Fold along breadth $\rightarrow$ halves the breadth

Each folded part:

- Length = l , Breadth = $\frac{b}{2}$

Perimeter:

$$2 \left(l + \frac{b}{2} \right) = 38 \Rightarrow 2l + b = 38 \quad (\text{Eq 2})$$

Solve the two equations:

From Eq 1:

$$l = 34 - 2b$$

Substitute into Eq 2:

$$\begin{aligned} 2(34 - 2b) + b &= 38 \\ 68 - 4b + b &= 38 \Rightarrow 68 - 3b = 38 \Rightarrow 3b = 30 \Rightarrow b = 10 \end{aligned}$$

Now find l :

$$l = 34 - 2(10) = 14$$

$$\text{Area} = l \times b = 14 \times 10 = \boxed{140 \text{ sq. cm}}$$



120-B

Solution

- Volumes of the three containers are in the ratio **3 : 4 : 5**.
- Ratios of **milk : water** in the containers:
 - First: 4 : 1
 - Second: 3 : 1
 - Third: 5 : 2

Let us assume the total volume of the three containers is a common multiple. Let it be:

- First container: 3 liters
- Second container: 4 liters
- Third container: 5 liters

Step 1: Calculate the amount of milk and water in each

First container (3 liters, 4:1 ratio)

Total parts = $4 + 1 = 5$

- Milk = $\frac{4}{5} \times 3 = 2.4$ liters
- Water = $\frac{1}{5} \times 3 = 0.6$ liters

Second container (4 liters, 3:1 ratio)

Total parts = 4

- Milk = $\frac{3}{4} \times 4 = 3$ liters
- Water = $\frac{1}{4} \times 4 = 1$ liter

Third container (5 liters, 5:2 ratio)

Total parts = 7

- Milk = $\frac{5}{7} \times 5 = \frac{25}{7} \approx 3.571$ liters
 - Water = $\frac{2}{7} \times 5 = \frac{10}{7} \approx 1.429$ liters
-

Step 2: Add all milk and water

- Total milk = $2.4 + 3 + \frac{25}{7} = \frac{168+210+250}{70} = \frac{628}{70} = 8.971$ liters
- Total water = $0.6 + 1 + \frac{10}{7} = \frac{42+70+100}{70} = \frac{212}{70} = 3.029$ liters

Step 3: Find ratio of milk : water

$$\text{Milk : Water} = \frac{628}{70} : \frac{212}{70} = 628 : 212$$

Simplify by dividing both by 4:

$$628 \div 4 = 157, \quad 212 \div 4 = 53$$

✅ **Final Answer:** The ratio of milk to water in the fourth container is **157 : 53**.

121-C

Solution

We are given:

- P and Q invest in the ratio 5 : 6 for a full year.
- R joins after 6 months and invests **an amount equal to Q's investment**.
- Total profit earned = Rs. 98,000 (which is **20%** of the capital).
- We need to find **R's investment**.

Step 1: Let's assume values for investment



Let:

- P's investment = $5x$
- Q's investment = $6x$
- R's investment = $6x$ (same as Q)

Now we calculate the **effective capital-time products** (investment $\times$ time):

- P: $5x \times 12 = 60x$
- Q: $6x \times 12 = 72x$
- R: $6x \times 6 = 36x$

Step 2: Ratio of profit shares

$$P : Q : R = 60x : 72x : 36x = 60 : 72 : 36$$

Divide each term by 12:

$$= 5 : 6 : 3$$

Step 3: Total profit = Rs. 98000

This is 20% of capital, so total capital =

$$\frac{98000}{0.2} = 490000$$



So, the profit is divided in the ratio $5 : 6 : 3 \rightarrow \text{Total parts} = 5 + 6 + 3 = 14$

R's share =

$$\frac{3}{14} \times 98000 = 21000$$

Now, R earned Rs. 21,000 from a **6-month** investment that resulted in 20% profit for the year.

So, Rs. 21,000 is **20% of 6x** (R's capital for 6 months):

Let amount invested by R be R_{amt} , and note that in 6 months, 20% annual profit translates to 10% profit.

$$10\% \text{ of } R_{amt} = 21000 \Rightarrow R_{amt} = \frac{21000 \times 100}{10} = \boxed{\text{Rs.210000}}$$

122-C

Here **liquid A** initially in the mixture is 7 parts out of 12 parts

The quantity of Liquid A will be 7 ka multiple (7,14,21,28.....)

Using option we can say quantity is **21**

Rest options are not a multiple of 7 so it can be eliminated.

123-C

We are given:

- Ratio of length to breadth of a rectangular park = $3 : 2$
- Speed of the cyclist = 12 km/hr
- Time taken to complete one round = 8 minutes
- We need to find the **area** of the park in m^2

Step 1: Convert speed and time to distance

$$\text{Speed} = 12 \text{ km/hr} = \frac{12 \times 1000}{60} = 200 \text{ m/min}$$

So in 8 minutes, the man covers:

$$200 \times 8 = 1600 \text{ m}$$

This is the **perimeter** of the park.



Step 2: Use perimeter formula

Let:

- Length = $3x$
- Breadth = $2x$

Then:

$$\text{Perimeter} = 2(l + b) = 2(3x + 2x) = 10x$$

Set this equal to 1600:

$$10x = 1600 \Rightarrow x = 160$$

So:

- Length = $3x = 480$ m
- Breadth = $2x = 320$ m

Step 3: Find the area

$$\text{Area} = \text{Length} \times \text{Breadth} = 480 \times 320 = \boxed{153600 \text{ m}^2}$$

Final Answer: The area of the park is 153600 square meters.

124-B

We are given:

- **A** invests Rs. 2000 for 12 months.
- **B** invests Rs. 3000 for 4 months, then Rs. 4000 for the remaining 8 months.
- **C** invests Rs. 4000 for 9 months, then Rs. 3000 for the last 3 months.
- Total profit = Rs. 8475.
- We are to find **A's share** of the profit.



Step 1: Calculate "investment months" (also called capital-time, t)

A:

$$\text{Capital} = 2000, \quad \text{Time} = 12 \Rightarrow \text{Capital-Time} = 2000 \times 12 = 24000$$

B:

- Rs. 3000 for 4 months $\rightarrow 3000 \times 4 = 12000$
- Rs. 4000 for 8 months $\rightarrow 4000 \times 8 = 32000$
- Total = $12000 + 32000 = 44000$

C:

- Rs. 4000 for 9 months $\rightarrow 4000 \times 9 = 36000$
- Rs. 3000 for 3 months $\rightarrow 3000 \times 3 = 9000$
- Total = $36000 + 9000 = 45000$

Step 2: Ratio of capital-time products

$$A : B : C = 24000 : 44000 : 45000$$

We simplify this ratio by dividing each term by 1000:

$$24 : 44 : 45$$

Step 3: Total parts = $24 + 44 + 45 = 113$

A's share =

$$\frac{24}{113} \times 8475 = \frac{203400}{113} = \boxed{\text{Rs.1800}}$$

125-C

Given:

- Triangle **ABC** has area = **20 sq. units**
- Lines **AP, BQ, CR** are drawn connecting midpoints, forming smaller triangles inside ABC
- You've noted:
 - **Area($\triangle APR$) = Area($\triangle PBQ$) = Area($\triangle BRC$) = Area($\triangle P\&R$)** (i.e., all small triangles formed are equal)

Goal:

To find the area of each smaller triangle, assuming the construction forms 4 equal  triangles within triangle ABC.

Since all 4 triangles are equal in area and total area is 20 sq. units, then:

$$\text{Area of each small triangle} = \frac{20}{4} = 5 \text{ sq. units}$$

126-B

Let speed of train = x m/s

Speed of man = 1.39 m/s

Relative speed = $x - 1.39$

$$\text{Distance} = \text{Relative Speed} \times \text{Time} \Rightarrow 125 = (x - 1.39) \times 10$$

$$x - 1.39 = \frac{125}{10} = 12.5 \Rightarrow x = 12.5 + 1.39 = \boxed{13.89 \text{ m/s}}$$

Final Answer:

$$\boxed{13.89 \text{ m/s}} \quad \text{or} \quad \boxed{50 \text{ km/h}} \quad (\text{after conversion})$$

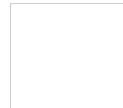
Conversion to km/h:

$$13.89 \times \frac{18}{5} = \boxed{50 \text{ km/h}}$$

127-C

- Length of train = 130 m
- Time to cross bridge = 30 seconds
- Speed = 45 km/h =

$$\frac{45 \times 1000}{3600} = 12.5 \text{ m/s}$$



Let the length of the bridge be x meters.

Total distance covered while crossing the bridge = length of train + length of bridge = $130 + x$

$$\text{Distance} = \text{Speed} \times \text{Time} \Rightarrow (130 + x) = 12.5 \times 30 = 375$$

$$130 + x = 375 \Rightarrow x = 375 - 130 = \boxed{245 \text{ meters}}$$

128-B

Two trains cross **each other in 23 seconds** while running in **opposite directions**.

- The **first train crosses a man in 27 seconds**.
- The **second train crosses a man in 17 seconds**.

If the **sum of the lengths of the two trains is 652 meters**, find their **speeds**.

Let's Assume:

Let:

- S_1 = Speed of train 1 (m/s)
- S_2 = Speed of train 2 (m/s)
- D_1 = Length of train 1
- D_2 = Length of train 2
- $D_1 + D_2 = 652 \text{ m}$

Step 1: Use Train–Man Equations

$$\text{Train 1 crosses man in 27 sec} \Rightarrow D_1 = S_1 \times 27 \quad \dots(\text{i})$$

$$\text{Train 2 crosses man in 17 sec} \Rightarrow D_2 = S_2 \times 17 \quad \dots(\text{ii})$$

Now add equations (i) and (ii):

$$D_1 + D_2 = 27S_1 + 17S_2 \quad \dots(\text{iii})$$

We also have:

$$\text{They cross each other in 23 sec:} \Rightarrow D_1 + D_2 = (S_1 + S_2) \times 23 \quad \dots(\text{iv})$$



Step 2: Equating (iii) and (iv):

$$27S_1 + 17S_2 = 23(S_1 + S_2)$$

Expand the right-hand side:

$$27S_1 + 17S_2 = 23S_1 + 23S_2$$

Bring like terms together:

$$27S_1 - 23S_1 = 23S_2 - 17S_2 \Rightarrow 4S_1 = 6S_2 \Rightarrow \frac{S_1}{S_2} = \frac{6}{4} = \frac{3}{2}$$

129-D

We are given:

- First person walks at 4.5 km/hr
- Second person walks at 5.4 km/hr
- Train takes:
 - 8.4 seconds to overtake the first person
 - 8.5 seconds to overtake the second person

Let the speed of the train be V km/hr and the length of the train be L meters.

Step 1: Convert speeds from km/hr to m/s

$$\text{Speed in m/s} = \text{Speed in km/hr} \times \frac{5}{18}$$

- Speed of first person = $4.5 \times \frac{5}{18} = 1.25$ m/s
- Speed of second person = $5.4 \times \frac{5}{18} = 1.5$ m/s
- Let train speed in m/s = $V \times \frac{5}{18}$

Step 2: Use the formula for distance = relative speed $\times$ time

Overtaking the first person:

$$L = (V \times \frac{5}{18} - 1.25) \times 8.4$$

Overtaking the second person:

$$L = \left(V \times \frac{5}{18} - 1.5\right) \times 8.5$$

Step 3: Equate the two equations for L

$$\left(V \times \frac{5}{18} - 1.25\right) \times 8.4 = \left(V \times \frac{5}{18} - 1.5\right) \times 8.5$$

Let $x = V \times \frac{5}{18}$, then:

$$(x - 1.25) \times 8.4 = (x - 1.5) \times 8.5$$

Expanding both sides:

$$8.4x - 10.5 = 8.5x - 12.75$$

Solving:

$$8.4x - 8.5x = -12.75 + 10.5$$

$$-0.1x = -2.25$$

$$x = \frac{2.25}{0.1} = 22.5$$

Now, recall $x = V \times \frac{5}{18}$

So,

$$22.5 = V \times \frac{5}{18} \Rightarrow V = \frac{22.5 \times 18}{5} = 81 \text{ km/hr}$$

130-B

Given:

A can do the work in **20 days** $\rightarrow$ A's 1-day work = $\frac{1}{20}$

B can do the work in **30 days** $\rightarrow$ B's 1-day work = $\frac{1}{30}$

C can do the work in **60 days** $\rightarrow$ C's 1-day work = $\frac{1}{60}$

A works **every day**

B and C assist **on every third day only**

So, work cycle:

- **Day 1:** A alone $\rightarrow$ work = $\frac{1}{20}$
- **Day 2:** A alone $\rightarrow$ work = $\frac{1}{20}$
- **Day 3:** A + B + C $\rightarrow$ work = $\frac{1}{20} + \frac{1}{30} + \frac{1}{60}$

Let's calculate the work done in every **3-day cycle**:

Work Done in 3 Days:

$$\text{Day 1} + \text{Day 2} = \frac{1}{20} + \frac{1}{20} = \frac{1}{10}$$

$$\text{Day 3} = \frac{1}{20} + \frac{1}{30} + \frac{1}{60} = \frac{3+2+1}{60} = \frac{6}{60} = \frac{1}{10}$$

So, in **3 days**, total work =

$$\frac{1}{10} + \frac{1}{10} = \frac{1}{5}$$

Work done in every 3 days = $\frac{1}{5}$

So, total time to complete 1 full work =

$$\frac{1}{1/5} = \boxed{5 \text{ days}}$$

131-A

Given:

- **6 men + 8 boys** can do the work in **10 days**
- **26 men + 48 boys** can do the same work in **2 days**
- We have to find the time taken by **15 men + 20 boys**

Step 1: Define Units of Work

Let:

- Work done by 1 man in 1 day = M
- Work done by 1 boy in 1 day = B



Step 2: Total Work

From the first condition:

$$(6M + 8B) \times 10 = \text{Total work} \quad \dots(i)$$

From the second condition:

$$(26M + 48B) \times 2 = \text{Total work} \quad \dots(ii)$$

Since both are equal to the same work:

$$(6M + 8B) \times 10 = (26M + 48B) \times 2$$

Now simplify:

$$60M + 80B = 52M + 96B$$

Bring all terms to one side:

$$60M - 52M = 96B - 80B \Rightarrow 8M = 16B \Rightarrow M = 2B$$

Step 3: Use this in total work

From (i):

$$(6M + 8B) \times 10 = \text{Work} \Rightarrow (6 \times 2B + 8B) \times 10 = (12B + 8B) \times 10 = 20B \times 10 = 200B$$

So, Total work = 200B

Step 4: Use new workforce (15 men + 20 boys)

$$15M + 20B = 15 \times 2B + 20B = 30B + 20B = 50B \quad (\text{work per day})$$

Final Step: Time = Work / Rate

$$\text{Time} = \frac{200B}{50B} = \boxed{4 \text{ days}}$$

132-D

Machine Rates (per hour):

- $P = 1/8 = 0.125$
- $Q = 1/10 = 0.1$
- $R = 1/12 \approx 0.0833$

From 9 A.M. to 11 A.M. (2 hours):

All three work together:

$$\text{Total Rate} = 0.125 + 0.1 + 0.0833 = 0.3083 \text{ lakh/hour}$$

In 2 hours:

$$\text{Work done} = 2 \times 0.3083 \approx 0.6166 \text{ lakh}$$

Remaining work:

$$1 - 0.6166 = 0.3834 \text{ lakh}$$

Now only Q and R are working:

$$\text{Combined rate of Q and R} = 0.1 + 0.0833 = 0.1833 \text{ lakh/hour}$$

Time to complete remaining:

$$\frac{0.3834}{0.1833} \approx 2.09 \text{ hours} \approx 2 \text{ hours and } 5 \text{ minutes}$$

Final Completion Time:

11 A.M. + 2 hours and 5 minutes $\approx$ 1:05 P.M.

But the question asks for "approximately" what time.

So rounding 1:05 P.M. to the nearest clean hour $\rightarrow$  1:00 P.M.

133-C

Given:

- A + B can do a piece of work in **8 days**
- B + C can do the same work in **12 days**
- A + B + C can do the same work in **6 days**

We are to find in how many days A and C together can complete the work.

Step 1: Assume Total Work = LCM of 8, 12, and 6 = 24 units

$$A + B \text{ efficiency} = \frac{24}{8} = 3 \text{ units/day}$$

$$B + C \text{ efficiency} = \frac{24}{12} = 2 \text{ units/day}$$

$$A + B + C \text{ efficiency} = \frac{24}{6} = 4 \text{ units/day}$$

Step 2: Find individual efficiencies

From:

$$\begin{aligned}(A + B + C) &= 4 \\(A + B) &= 3 \Rightarrow C = 4 - 3 = 1 \\(B + C) &= 2 \Rightarrow A = 4 - 2 = 2 \\&\Rightarrow A = 2, C = 1\end{aligned}$$

So,

- A's efficiency = 2 units/day
- C's efficiency = 1 unit/day
- $\Rightarrow A + C = 3$ units/day

Step 3: Time = Total Work / Efficiency

$$\text{Time} = \frac{24}{3} = \boxed{8 \text{ days}}$$

134-A

Given:

- The **sum of ages of 5 children is 50 years**
- They were **born at intervals of 3 years**
- Let the **youngest child's age = x**

So, the children's ages will be:

- x (youngest)
- $x + 3$
- $x + 6$
- $x + 9$
- $x + 12$

Step 1: Write the equation

Sum of their ages:

$$x + (x + 3) + (x + 6) + (x + 9) + (x + 12) = 50$$

Add terms:

$$5x + (3 + 6 + 9 + 12) = 50 \Rightarrow 5x + 30 = 50 \Rightarrow 5x = 20 \Rightarrow x = \boxed{4}$$

The **youngest child is 4 years old.**

135-A

Given:

- Present age of **Sameer** = $5x$
- Present age of **Anand** = $4x$
- After **3 years**, their ages are in the **ratio 11:9**

So:

$$\frac{5x + 3}{4x + 3} = \frac{11}{9}$$

Step 1: Cross Multiply

$$9(5x + 3) = 11(4x + 3)$$

$$45x + 27 = 44x + 33$$

Now solve:

$$45x - 44x = 33 - 27 \Rightarrow x = 6$$

Step 2: Find Ages

$$\text{Sameer's present age} = 5x = 5 \times 6 = 30$$

$$\text{Anand's present age} = 4x = 4 \times 6 = 24$$

136-B

Given:

- Present ages = $4x, 7x, 9x$
- **8 years ago**, their ages were:
 - $4x - 8$
 - $7x - 8$
 - $9x - 8$

$$\text{The sum of their ages 8 years ago} = 56$$

Step 1: Equation

$$(4x - 8) + (7x - 8) + (9x - 8) = 56$$

$$(4x + 7x + 9x) - 24 = 56 \Rightarrow 20x - 24 = 56 \Rightarrow 20x = 80 \Rightarrow x = 4$$

Step 2: Present Ages

$$4x = 16, \quad 7x = 28, \quad 9x = 36$$

Present ages are:

16 years, 28 years, and 36 years

137-C

We'll consider all valid combinations where the number of men is **at least 3**. That means:

- 3 men + 2 women
- 4 men + 1 woman
- 5 men + 0 women

We'll use combinations for counting.

Case-wise Breakdown:

Case 1: 3 men + 2 women

$$\binom{7}{3} \times \binom{6}{2} = 35 \times 15 = 525$$

Case 2: 4 men + 1 woman

$$\binom{7}{4} \times \binom{6}{1} = 35 \times 6 = 210$$

Case 3: 5 men + 0 women

$$\binom{7}{5} \times \binom{6}{0} = 21 \times 1 = 21$$

Total Ways:



$$525 + 210 + 21 = \boxed{756}$$

Final Answer:

756 ways

Note: () means nCr

Formula is $= \frac{n!}{r!(n-r)!}$

138-A

Step 1: Count the letters

The word **LEADER** has **6 letters**.

But — some letters repeat.

☐ L: 1

☐ E: **2** (repeated)

☐ A: 1

☐ D: 1

☐ R: 1

Step 2: Apply the formula

When **letters** are repeated, the number of distinct arrangements is:

$$\frac{n!}{(\text{repetition factorials})}$$

Here:

$$\frac{6!}{2!} = \frac{720}{2} = \boxed{360}$$

360 distinct arrangement

139-C

Step-by-Step Logic:

Total letters = 10 (L, O, G, A, R, I, T, H, M, S)

All letters are **distinct**

We want to form **4-letter arrangements**, so this is a **permutation** problem (order matters)

$$\text{Number of 4-letter words} = P(10, 4) = 10 \times 9 \times 8 \times 7 = \boxed{5040}$$

Formula Used:

$$P(n, r) = \frac{n!}{(n - r)!}$$

Where:

- n = total number of distinct items (letters)
- r = number of items to arrange (letters in the word)

For this case:

$$P(10, 4) = \frac{10!}{(10 - 4)!} = \frac{10!}{6!} = 10 \times 9 \times 8 \times 7 = 5040$$

140-B

Cube root of 0.000216 is **0.06**

141-A

Given:

Distance $D = 600$ km

Speed reduced by **200 km/h**

Time increased by **30 minutes** = $\frac{1}{2}$ hour

Let original speed be x km/h

New speed = $x - 200$

Formula used:



$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\frac{600}{x - 200} - \frac{600}{x} = \frac{1}{2}$$

Solve:

Multiply both sides by $x(x - 200)$:

$$600x - 600(x - 200) = \frac{1}{2}x(x - 200)$$

$$600x - 600x + 120000 = \frac{1}{2}x(x - 200)$$

$$120000 = \frac{1}{2}x(x - 200)$$

$$240000 = x(x - 200)$$

$$x^2 - 200x - 240000 = 0$$

Solve this quadratic:

$$x = \frac{200 \pm \sqrt{200^2 + 4 \times 240000}}{2} = \frac{200 \pm \sqrt{1000000}}{2} = \frac{200 \pm 1000}{2}$$

$$x = 600 \text{ or } -400$$

Take $x = 600$

original speed was 600, and flight happened **before** slowdown.

$$\frac{600}{600} = \boxed{1 \text{ hour}}$$

142-B

Given:

- Height of lighthouse = 100 m
- Angles of elevation from the two ships = 30° and 45°
- Ships are on **opposite sides** of the lighthouse
- Need to find the **distance between the two ships**
- **Use of Trigonometry:**
- Let:

- Distance of ship 1 from lighthouse (angle 30°) = x
- Distance of ship 2 from lighthouse (angle 45°) = y



From trigonometry:

- $\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$

So,

$$1. \tan(30^\circ) = \frac{100}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{100}{x} \Rightarrow x = 100\sqrt{3}$$

$$2. \tan(45^\circ) = \frac{100}{y} \Rightarrow 1 = \frac{100}{y} \Rightarrow y = 100$$

Total Distance between the two ships:

$$\text{Distance} = x + y = 100\sqrt{3} + 100 = 100(\sqrt{3} + 1)$$

$$\text{Distance} \approx 100(1.732 + 1) = 100(2.732) = \boxed{273.2 \text{ m}}$$

Final Answer:

$$\boxed{273.2 \text{ meters}}$$

143-C

We are given:

- Length of pipe $h = 21 \text{ cm}$
- External diameter $D = 8 \text{ cm} \Rightarrow$ External radius $R = 4 \text{ cm}$
- Thickness = $1 \text{ cm} \Rightarrow$ Internal radius $r = 4 - 1 = 3 \text{ cm}$
- Density of iron = 8 g/cm^3

We need to find the **weight** of the pipe, which is:

Step 1: Volume of the pipe (hollow cylinder)

$$\text{Volume} = \pi h(R^2 - r^2)$$

Substitute the values:

$$= \pi \cdot 21 \cdot (4^2 - 3^2) = \pi \cdot 21 \cdot (16 - 9) = \pi \cdot 21 \cdot 7 = 147\pi \text{ cm}^3$$

$$\begin{aligned}\text{Volume} &= 147\pi \text{ cm}^3 \\ \text{Weight} &= 147\pi \times 8 = 1176\pi \text{ grams}\end{aligned}$$



Using a more accurate value of $\pi = 3.14159$:

$$1176 \times 3.14159 = 3696.01584 \text{ grams} \approx 3696 \text{ grams} = 3.696 \text{ kg}$$

144-C

Given:

$$\text{Time upstream} = 8 \text{ hrs } 48 \text{ mins} = 8 + \frac{48}{60} = 8.8 \text{ hours}$$

$$\text{Time downstream} = 4 \text{ hours}$$

Distance is the same in both directions

Let:

- Speed of boat in still water = b
- Speed of current = c

Then:

- Upstream speed = $b - c$
- Downstream speed = $b + c$

Let the distance be D . Then:

$$\frac{D}{b - c} = 8.8 \quad (1)$$

$$\frac{D}{b + c} = 4 \quad (2)$$

Now, divide (1) by (2):

$$\frac{8.8}{4} = \frac{b+c}{b-c} \Rightarrow 2.2 = \frac{b+c}{b-c}$$



Cross-multiply:

$$\begin{aligned} 2.2(b-c) &= b+c \\ 2.2b - 2.2c &= b+c \end{aligned}$$

Bring like terms together:

$$2.2b - b = 2.2c + c \Rightarrow 1.2b = 3.2c \Rightarrow \frac{b}{c} = \frac{3.2}{1.2} = \frac{32}{12} = \frac{8}{3}$$

145-A

Given:

Distance = 36 miles

Speed of boat in still water = 10 mph

Let speed of stream = x mph

Downstream speed = $10 + x$

Upstream speed = $10 - x$

Time difference = 90 minutes = 1.5 hours

Time taken:

$$\text{Time downstream} = \frac{36}{10+x}$$

$$\text{Time upstream} = \frac{36}{10-x}$$

Given that upstream time is more by 1.5 hours:

$$\frac{36}{10-x} - \frac{36}{10+x} = 1.5$$

Solve the equation:

Take LCM:



$$36 \left(\frac{1}{10-x} - \frac{1}{10+x} \right) = 1.5$$

Use identity:

$$\frac{1}{a-b} - \frac{1}{a+b} = \frac{2b}{a^2 - b^2}$$

So:

$$36 \cdot \frac{2x}{100 - x^2} = 1.5$$

$$\frac{72x}{100 - x^2} = 1.5$$

Multiply both sides by $100 - x^2$:

$$72x = 1.5(100 - x^2)$$

$$72x = 150 - 1.5x^2$$

Multiply all terms by 2 to eliminate decimals:

$$144x = 300 - 3x^2 \Rightarrow 3x^2 + 144x - 300 = 0$$

Divide by 3:

$$x^2 + 48x - 100 = 0$$

Use quadratic formula:

$$x = \frac{-48 \pm \sqrt{48^2 + 4 \cdot 100}}{2} = \frac{-48 \pm \sqrt{2304 + 400}}{2} = \frac{-48 \pm \sqrt{2704}}{2} = \frac{-48 \pm 52}{2}$$

Take the positive root:

$$x = \frac{-48 + 52}{2} = \frac{4}{2} = 2$$

Given:

Distance = $\frac{3}{4}$ km = 0.75 km

Time upstream = 11 min 15 sec = $\frac{45}{4}$ min = $\frac{45}{240}$ hr = $\frac{3}{16}$ hr

Time downstream = 7 min 30 sec = $\frac{15}{2}$ min = $\frac{15}{120}$ hr = $\frac{1}{8}$ hr

Speed = Distance / Time

Upstream speed = $\frac{0.75}{3/16} = 0.75 \times \frac{16}{3} = 4$ km/hr

Downstream speed = $\frac{0.75}{1/8} = 0.75 \times 8 = 6$ km/hr

Speed of man in still water:

$$\text{Speed in still water} = \frac{4 + 6}{2} = \boxed{5 \text{ km/hr}}$$

147-B

We are given:

- Pipe A fills the tank in **37 minutes 30 seconds = 37.5 minutes**
- Pipe B fills the tank in **45 minutes**
- Both pipes are opened together, but **B is turned off after some time**
- Total time to fill the tank = **30 minutes**
- We need to find **after how much time pipe B is turned off**

Step 1: Work rates

Let the total work (i.e., filling the tank) be 1 unit.

- Rate of A = $\frac{1}{37.5}$ units/minute
- Rate of B = $\frac{1}{45}$ units/minute

Let B be turned off after x minutes.

So:

- For x minutes, both A and B work
- For remaining $30 - x$ minutes, only A works

Step 2: Total work done

$$\text{Work} = x \left(\frac{1}{37.5} + \frac{1}{45} \right) + (30 - x) \left(\frac{1}{37.5} \right) = 1$$

Factor and simplify:

$$x \left(\frac{1}{37.5} + \frac{1}{45} - \frac{1}{37.5} \right) + 30 \left(\frac{1}{37.5} \right) = 1$$

$$x \left(\frac{1}{45} \right) + \frac{30}{37.5} = 1$$

$$\frac{x}{45} + \frac{30}{37.5} = 1$$

$$\frac{x}{45} + 0.8 = 1 \Rightarrow \frac{x}{45} = 0.2 \Rightarrow x = 9$$

Pipe B is turned off after 9 minutes.

148-C

Given:

Pipe A fills the tank in 15 minutes $\rightarrow$ efficiency = $\frac{1}{15}$

Pipe B fills the tank in 20 minutes $\rightarrow$ efficiency = $\frac{1}{20}$

Both are opened together for 4 minutes, then A is turned off.

Step 1: Work done in first 4 minutes

$$\text{Combined rate of A and B} = \frac{1}{15} + \frac{1}{20} = \frac{4+3}{60} = \frac{7}{60}$$

$$\text{Work done in 4 minutes} = 4 \times \frac{7}{60} = \frac{28}{60}$$

Step 2: Remaining work

$$\text{Remaining work} = 1 - \frac{28}{60} = \frac{32}{60} = \frac{8}{15}$$

Step 3: Now only B is working (rate = 1/20)

$$\text{Time taken by B to do } \frac{8}{15} \text{ work} = \frac{8}{15} \div \frac{1}{20} = \frac{8}{15} \times 20 = \frac{160}{15} = \frac{32}{3} = 10\frac{2}{3} \text{ minutes}$$

Total time = First 4 minutes + $10\frac{2}{3}$ minutes =

$$14\frac{2}{3} \text{ minutes (or 14 minutes 40 seconds)}$$

149-D

Formula:

$$\theta = \left| \left(30H - \frac{11M}{2} \right) \right|$$

Where:

- H = hour
- M = minutes
- θ = angle between the hands

Substitute values ($H = 4$, $M = 20$):

$$\theta = \left| 30 \times 4 - \frac{11 \times 20}{2} \right| = |120 - 110| = |10| = 10^\circ$$

150-A

We are given:

- A and B run a 100 m race.
- A runs at **5 km/h**.
- A gives B a **start of 8 m**, so B runs only **92 m**.
- A **still beats B by 8 seconds**.
- We are to find the **speed of B**.

Step 1: Convert A's speed to m/s

$$5 \text{ km/h} = \frac{5 \times 1000}{60 \times 60} = \frac{5000}{3600} = \frac{25}{18} \text{ m/s}$$

Step 2: Time taken by A to run 100 m

$$\text{Time}_A = \frac{100}{25/18} = 100 \times \frac{18}{25} = 72 \text{ seconds}$$

Step 3: Time taken by B = A's time + 8 sec = 72 + 8 = **80 seconds**

Distance run by B = 92 m

$$\text{Speed}_B = \frac{92}{80} = 1.15 \text{ m/s}$$

Convert to km/h:

$$1.15 \times \frac{18}{5} = 4.14 \text{ km/h (approximately)}$$

151-C

ADBCEF

A is the introductory statement and F is the conclusion statement

152-A

ACEDBF makes a perfect coherent paragraph

153-C

BEADC

154-D

ECBDA

155-B

BEDCA

156-C

Obviously perfectly fits here

157-A

Quantitatively

158-D
Alternatives

159-A
Resolve

160-D
Allowed

161-A
Synonyms – calmness , reticent , silent

162-B
Synonym – make thin

163-C
Synonym – imperious
Meaning - When you're haughty, you have a big attitude and act like you're better than other people

Other options
Cemetery- a large burial ground, especially one not in a churchyard.
Adamant- refusing to be persuaded or to change one's mind

164-B
Synonym – retribution
Other options
Indulgence - the attitude of allowing yourself or someone else to have something enjoyable, or the act of having something enjoyable

165-D
Synonym – disdain
Deride- Some common synonyms of deride are mock, ridicule, and taunt

166-C
Antonym – absurdity
Scenic- providing or relating to views of impressive or beautiful natural scenery.
Subtle- (especially of a change or distinction) so delicate or precise as to be difficult to analyse or describe.

167-A
Antonym – emphasize

168-C
Antonym – endorse

169-C

One word substitution is – Dotage

Serendipity- the occurrence and development of events by chance in a happy or beneficial way.

Superannuation- Superannuation refers to an organized retirement savings system where regular contributions are made into a fund during an individual's working life

Ethnocity- the quality or fact of belonging to a population group or subgroup made up of people who share a common cultural background or descent

170-B

One word substitution for this is – fatalist

Tyrant- cruel ruler

Puritan- a member of a group of English Protestants of the late 16th and 17th centuries who regarded the Reformation of the Church under Elizabeth I as incomplete and sought to simplify and regulate forms of worship.

Egalitarian- believing in or based on the principle that all people are equal and deserve equal rights and opportunities.

171-C

If the villagers did not **get** him admitted - contains error , sentence is in the past form

So it should be **got** instead of get in the sentence

172-D

And **then** went down his doom very quickly - contains error

It should be **than** instead of then. Bcz here there is a comparison between honorary and doom days

173-B

Most interesting book - contains error , it should be **books** not book

Because after using **one of the** _____.

Here **plural subject** will come in place of blank

174-A

35214

Makes a perfect sequence

175-D

52431 forms the correct sequence

176-C

Meaning is – to confess without reserve.

177-B

To catch a tartar" is an idiom that means to encounter someone or something that is unexpectedly difficult, challenging, or troublesome



178-D

Meaning - to have a strong personal opinion about something that you want people to accept and that is the reason why you do something

Answer is a private end to serve.

179-C

Meaning – to criticize someone

180-B

Meaning find fault with a gift

181-D

Lincoln initially wanted **emancipation to be a slow and gradual process**, allowing time for society to adjust. He even proposed compensating slave-owners and colonization plans for rehabilitating freed slaves. However, his plan was disrupted when the **Southern states hastily decided to secede from the Union**, leading to the **Civil War**. The urgency and pressure of war **forced Lincoln to act quickly**, abandoning his gradual approach in favor of immediate emancipation to preserve the Union and counter the rebellion.

182-A

By **freeing the slaves**, Lincoln believed they were also **protecting and strengthening the freedom of all citizens**, including those who were already free. It emphasizes that **true liberty for a nation** can only exist when **everyone within it is free**—ensuring justice and moral integrity for the whole country.

183-B

In the passage, the term "**colonization**" refers to **Abraham Lincoln's proposal to resettle freed Black people outside the United States**, often to places like Africa or Central America.

He believed this might help with the **rehabilitation of freed slaves** by giving them a new start elsewhere, as he thought integration into American society would be difficult due to deep-rooted racial prejudices at the time.

184-E

His way of gradual emancipation

185-C

Hell bent **upon** getting is the correct usage it should be **upon** not be at getting

186-C

Had the room been brighter is the answer

Because in a particular sentence. Sentence cannot have a condition of future tense and also having future work like (reading , writing etc.) to be done

187-C

188-D

Logic - The relationship between "indigent" and "wealthy" is one of opposition or . "Indigent" means poor or needy, while "wealthy" means rich or possessing a great deal of financial resources. These words represent the opposite ends of the socioeconomic spectrum.

189-B

The spiritual and corporeal realms are often used in an analogous way to represent the inner and outer aspects of human existence. The corporeal realm refers to the physical body and its tangible experiences, while the spiritual realm encompasses the intangible aspects like consciousness, emotions, and beliefs.

190-C

They know what it requires a long

What it requires is wrong usage it should be **that it takes**

Explanation of passage (191 to 194)

Short Explanation of the Passage on Robots:

1. **History of Robots:**

The word "robot" was first used in a 1921 play by Karel Čapek. Real robots began being developed in the 1960s by engineers like Joseph Engelberger and George Devol.

2. **Use in Industry:**

Early robots helped in factories—especially in dangerous tasks. They didn't tire, eat, or get distracted.

3. **Advanced Applications:**

Robots are now used for risky jobs like bomb disposal, volcano research, space missions (e.g., Mars Rovers).

4. **Household Use:**

Robots like *Roomba* are used in homes for cleaning.

5. **Future of Robots:**

In 2005, a robot car named *Stanley* completed a desert race without a driver, showing potential for driverless cars.

6. **Conclusion:**

Robots have evolved from scary science fiction to real-world helpers in homes, industries, and research. In the future, they will take over more human tasks.

191-D

In the future people wont drive cars – robot will

192-A

The first real robot was installed in an automobile plant

193-B

Industrial robots perform jobs that require high heat

194-A

In industry , Robots

Have taken away some jobs and created others

195-A

His father said go to your room and study



196-C

The passage highlights the **contradictory nature of the Indian middle class**. Though it deeply **abhors corruption**, it is also widely involved in it. Over time, **shock over corruption has declined** due to growing **material ambitions** and **limited opportunities**. Despite repeated disappointments, this class continues to **strongly yearn for a corruption-free society** and continues to support those who promise to fight corruption.

197-B

Teachers are not high on the list of corruption because they do not have **Opportunities**

198-D

The Indian middle class is **stratified**

Stratified means divided into levels or classes based on social status, education, or income.

199-D

In the passage, the phrase "**God in man**" implies the **realization of a divine or higher spiritual essence within human beings**. It suggests that true salvation and transformation do not come from external systems like politics or economics, but from **awakening a deeper moral and spiritual consciousness**—characterized by **love, inner growth, and self-awareness**. This inner realization or awakening is referred to as recognizing or realizing the presence of **God within oneself**.

200-A

Based on the passage, the author would tend to agree with the statement that:

"Efforts to educate young women have significantly contributed to the decline in global illiteracy rates."

This is supported by the passage mentioning that the **sharp increase in literacy rates among young women**—a result of targeted **educational campaigns for girls**—is one of the **major reasons** for the decline in worldwide illiteracy.